TEACHER PACK

Grade 6 (Year 7) | Phase 3 / Cycle 2 | AI Solution Builder

Main Course, Term 1 - Rule-Based vs Learning Systems (ML Foundations)

Emerging tier | 10 weeks (Core + Lab)

Field	Detail
Program	Grade 6 Main Course, Term 1 (Emerging tier)
Unit theme	Rule-Based vs Learning Systems - ML Foundations
Signature activity	A structured, prompt-driven AI solution to a simple task, evaluated for accuracy and risk
Strands	AI Conceptual Understanding | AI Solution Design & Development | Critical Evaluation & Informed Interaction | Ethics & Governance
Delivery model	2 x 45-minute periods per week - a Core lesson + a paired Lab lesson each week
Lessons in this pack	20 lessons (10 Core + 10 Lab)

1. About this Teacher Pack

This pack contains everything needed to teach the Grade 6 main course, Term 1: ten weeks of Emerging-tier lessons in which students distinguish rule-based from learning systems, meet machine-learning foundations (training and testing data), use structured prompts to complete tasks, judge outputs, weigh risks, and build a structured, prompt-driven AI solution evaluated for accuracy and risk. It opens Phase 3 (AI Solution Builder) and steps up from Grade 5 in technical depth, cognitive demand, and safeguarding. Each week is a Core lesson (teaches the outcome) and a paired Lab lesson (deeper hands-on build/investigation on the same outcome). Weeks 1-5 teach one strand focus each; Week 6 integrates them; Weeks 7-10 plan, build, refine and showcase the signature solution.

How to use this pack

- Teach each week as Core (period 1) then Lab (period 2) on the same outcome, across the term.
- Keep all AI/ML use teacher-mediated and account-free; train any model only on non-personal data (objects/drawings/sounds); never enter personal data.
- Build the evaluate-verify-risk habit into every AI activity; provide the unplugged fallback whenever devices or approval are unavailable.
- Use the per-lesson formative check to confirm the outcome before moving on.
- Pair this Teacher Pack with the Student Lesson Content and Student Worksheets (same order).
- Weeks 7-10 run the signature solution end-to-end: plan (approve-to-build gate), build, refine, showcase and assess.
- Strand-naming note: the ethics strand is labelled 'Responsible AI Use' (Grade 4), 'Ethics & Responsible Use' (Grade 5), and 'Ethics & Governance' (Grade 6); the data strand is labelled 'Data Awareness' (Grade 4), 'Data Literacy' (Grade 5), and folded into AI Conceptual Understanding from Grade 6 onward - these track the same strand across grades, not separate strands.

Safeguarding & responsible use

Grade 6 adds introductory ML on top of AI interaction. Keep all AI and ML use teacher-mediated and account-free for students; prefer board demonstrations, simulated-AI cards, or curated output sets, or a teacher-managed school-approved child-safe assistant. Train any classifier only on non-personal data - objects, drawings, shapes, or sounds - never faces, voices, names, or any personal or sensitive data. Never enter personal data into a prompt. Build evaluation, verification, and risk checks into every activity. Keep all activity supervised, use only school-approved devices, post nothing online, and follow ADEK safeguarding and responsible-use guidance. Contexts and names are UAE-appropriate and inclusive.

Materials & tools (UAE-accessible)

Tool / material	What it is & why	Access route	Unplugged fallback
Rule vs learning demos + system cards (unplugged)	A rule-card machine and a learn-from-examples card game, plus real-system sort cards (W1-W2).	School-printed; teacher-run; no account.	Discuss familiar systems and sort on the board.
Teachable classifier (introductory ML, teacher-mediated)	An age-appropriate image/sound/pose classifier trained on NON-personal data (objects, drawings, shapes, sounds) to show how a model learns (W2).	Browser-based, teacher-managed device; no student accounts; no faces/voices/names.	Unplugged paper-classifier: students hand-sort training and testing cards.
Simulated-AI / teacher-mediated AI demo	Board demonstrations and simulated-AI cards for structured prompting, task completion, and output triage (W3-W9).	Account-free; teacher-run; no personal data.	Curated pre-generated output sets.

Output-evaluation, verification & risk sheets (printable)	Triage cards, reliability checklists, verification flowcharts, and a risk matrix (W4-W10).	School-printed.	Record in an exercise book.
Signature-solution planner & rubric (printable)	Solution-architecture planner, project success criteria, build/test logs, and the Emerging analytic rubric (W6-W10).	School-printed.	Plan and log on paper.
School-approved AI assistant (optional, teacher-managed)	A child-safe, school-approved assistant used only by the teacher on the board for a live demo, if permitted.	Teacher-managed class account; no student sign-in; no personal data.	Simulated-AI cards or curated output sets.

Program at a glance

#	Lesson	Strand	Framework ref(s)
Week 1 - Core	Unit Launch - Rule-Based vs Learning Systems	Integration	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 1 - Lab	Rule vs Learning - Hands-On Exploration (Lab)	Integration	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 2 - Core	Rules vs Learning - How Each System Works	AI Conceptual Understanding	AIF·CU·G6·E
Week 2 - Lab	Train a Simple Model (Lab)	AI Conceptual Understanding	AIF·CU·G6·E
Week 3 - Core	Structured Prompts to Complete a Task	AI Solution Design & Development	AIF·SD·G6·E
Week 3 - Lab	Prompt-to-Task Workshop (Lab)	AI Solution Design & Development	AIF·SD·G6·E
Week 4 - Core	Judging Outputs - Correct, Incorrect, Unclear	Critical Evaluation & Informed Interaction	AIF·CE·G6·E
Week 4 - Lab	Output Triage Clinic (Lab)	Critical Evaluation & Informed Interaction	AIF·CE·G6·E
Week 5 - Core	Risks and Responsible Use	Ethics & Governance	AIF·GE·G6·E
Week 5 - Lab	Class AI Risk Audit (Lab)	Ethics & Governance	AIF·GE·G6·E
Week 6 - Core	Bringing the Strands Together	Integration	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 6 - Lab	Integrated Build-and-Check (Lab)	Integration	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 7 - Core	Signature Solution - Planning	Integration (Signature Solution - Planning)	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 7 - Lab	Prototype and Pressure-Test the Plan (Lab)	Integration (Signature Solution - Planning)	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E

Week 8 - Core	Signature Solution - Building	Integration (Signature Solution - Building)	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 8 - Lab	Extended Build and Evaluation (Lab)	Integration (Signature Solution - Building)	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 9 - Core	Signature Solution - Refining & Finishing	Integration (Signature Solution - Refining & Finishing)	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 9 - Lab	Final Iteration and Risk Review (Lab)	Integration (Signature Solution - Refining & Finishing)	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 10 - Core	Showcase, Assessment & Reflection	Integration (Showcase, Assessment & Reflection)	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E
Week 10 - Lab	Showcase Carousel and Reflection (Lab)	Integration (Showcase, Assessment & Reflection)	AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E

Week 1 - Core | Unit Launch - Rule-Based vs Learning Systems

Strand: Integration | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Core)

Objectives & success criteria

- Generate at least two questions about how rule-based and learning systems differ.
- Classify three systems as 'follows fixed rules' or 'learns from data', justifying each.
- Co-construct the unit success criteria and the responsible-use norms for AI tools.

Student-facing success criteria:

- I can ask at least two genuine questions about rule-based and learning systems.
- I can sort three systems into 'follows fixed rules' or 'learns from data' and give a reason for each.
- I can help write our class responsible-use rules for AI tools.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you explored what AI can and cannot do and practised giving clear instructions to a machine. This unit builds on that by asking a deeper question: does the machine follow fixed rules a person wrote, or does it learn patterns from examples? You do not need any coding for today.

Activate: Show two short board clips or cards: a vending machine giving change (fixed steps) and a photo app grouping pictures of cats. Ask: which one was told exactly what to do, and which one seems to have figured something out?

Teach: Introduce the big idea of the whole unit - some systems follow rules written by people, and some systems learn patterns from data. Give the plain definitions and one example of each.

Apply: In pairs, students place three system cards (traffic light timer, spam-email filter, spell-checker suggestions) under 'fixed rules' or 'learns from data' and write one reason.

Check: Each pair reads out one sorted card and its reason; you note which distinctions are secure and which are shaky for the KWL.

Critical & analytical thinking focus

Question & analyse - frame the rule-vs-learning distinction (Understand/Analyse | DOK 1-2 | SOLO: Uni to Multistructural)

The technical concept (teacher background)

A rule-based system does exactly what a person has written into it, step by step. If a situation is not covered by a rule, the system cannot handle it, because it has no rules to follow. A learning system is different - instead of being given every rule, it is shown many examples and it finds patterns in that data. We call the pattern it builds a model. The same learning system can behave differently if you train it on different examples, because what it learns depends on the data it sees. This unit is the start of Phase 3, where you begin to think like an AI solution builder, so you need to tell these two kinds of system apart. Neither kind is magic - both are built and checked by people. Understanding which kind you are looking at helps you predict when a system will work well and when it might fail.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Hook and baseline 8 min	Show the vending machine and photo-grouping clips; hand out the baseline KWL sheet and ask students to write what they already Know and Want to know.	Watch, then jot one 'know' and one 'want to know' about how machines make decisions.	• Which machine was told every step? • Which one seems to have worked something out for itself?
Teach the big idea 10 min	Define rule-based system and learning system with the comparison visual; model one everyday example of each and think aloud about the difference.	Follow along, add the two definitions and one example each to the student page.	• Where do the rules come from in each case? • What does a learning system need that a rule-based one does not?
Sort three systems	Hand pairs the three system cards; circulate and press for reasons, not	Sort traffic-light timer, spam filter and spell-checker into	• What clue tells you it learned from examples?

12 min	guesses.	fixed-rules or learns-from-data; write a reason for each.	• Could a person write every rule for this task?
Question wall 8 min	Invite pairs to post at least two genuine questions on the class question wall; group similar questions live.	Write and post two questions you still have about rule-based versus learning systems.	• What is still puzzling about how a system learns? • Which question would help us most this unit?
Co-construct norms 7 min	Lead a short discussion to agree the unit success criteria and a responsible-use charter (teacher-only tools, no personal data, always check outputs).	Suggest one class rule for using AI safely and fairly; record the agreed charter on the student page.	• Why should we never type someone's name or photo into an AI tool? • How will we check if an AI output is correct?

Anticipated misconceptions

Misconception	How to address it
A learning system just follows a big set of fixed rules that a programmer typed in.	Show that no person wrote a rule for every cat photo. The system was shown many examples and found the pattern itself. That pattern is the model, and it can change if the examples change.
If a machine gives an answer, it must understand the topic like a person.	Explain that a learning system matches patterns in data; it has no understanding or feelings. A confident-looking answer can still be wrong, which is why we always check.
Rule-based systems are old and useless, and learning systems are always better.	Both are useful. A traffic-light timer works perfectly with simple rules. Learning systems are only better when the task is too varied for a person to write every rule.

Hands-on activity guide

Overview: An unplugged sorting and questioning launch: students compare a rule-card machine (fixed steps) with a set of learn-from-examples cards, then sort three real systems and post questions. No devices or accounts are used by students.

Grouping / timing: Pairs for sorting, whole class for the question wall and charter.

Tool: Printed system cards and the baseline KWL sheet (teacher board demo only for the two clips). - Account-free for students. Any AI clip is a curated, teacher-shown board demo using non-personal examples only.

Facilitation steps:

1. Watch the two short board clips and note which machine was told every step.
2. Read the two definitions and add one example of each to your page.
3. Sort the three system cards into 'fixed rules' or 'learns from data' with your partner.
4. Write one reason under each card - what is your evidence?
5. Post at least two genuine questions on the class question wall.
6. Agree one responsible-use rule and add it to the class charter.

Unplugged fallback: If no board clips are available, read the two printed scenario cards aloud and sort using only the printed cards - the lesson runs fully unplugged.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Baseline for the rule-based vs learning-system distinction and safe-use norms

• Check: Diagnostic KWL + question wall + rule/learning sort (not graded)

• Success criteria: Poses at least 2 genuine questions and sorts at least 2 of 3 systems with a reason.

• Evidence: Sorted systems + question wall on the baseline sheet.

• Feedback / adjustment: Cluster misconceptions to target Weeks 2-5 and plan the Primer; agree the responsible-use charter.

Differentiation

Representation: Pair each definition with an icon and a real photo-free example; colour-code the two sort columns.

Action & expression: Allow sorting by placing physical cards, pointing, or speaking the answer to a partner.

Engagement: Use familiar UAE examples (Salik toll gates as rule-based, a photo app grouping desert and beach pictures as learning).

People of Determination: Provide pre-cut large-print cards and a two-symbol choice board (gear = rules, seed = learns) for students who find writing hard; allow a scribe.

EAL: Give a bilingual glossary card for rule, learn, data, model, example; sentence frame 'This is ___ because ___'.

Stretch / next tier: Ask students to invent a fourth system card and challenge another pair to classify and justify it.

Answer key / model responses

Task / item	Model response	Marking note
Sort the traffic-light timer.	Fixed rules - it changes lights on a set timer that a person programmed.	Accept 'learns' only if the student describes a smart signal that adapts to traffic data.
Sort the spam-email filter.	Learns from data - it was trained on many example emails marked spam or not spam.	Key clue: it improves from examples, and no person wrote a rule for every email.
Write one question for the wall.	Example: 'How many examples does a system need before it learns well?'	Any genuine, on-topic question counts; reward curiosity over neatness.
Suggest a responsible-use rule.	Example: 'We never type a real person's name, photo or voice into an AI tool.'	Look for a rule about privacy or checking outputs.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Classifying systems	Sorts at least 2 of 3 systems correctly with a sensible reason.	Sorts 1 system correctly or sorts without clear reasons.	Cannot yet sort or gives unrelated reasons.
Questioning	Poses at least two genuine, on-topic questions.	Poses one on-topic question.	No question posed yet.
Responsible-use thinking	Suggests a clear safe-use rule and explains why.	Suggests a rule with little reason.	No safe-use idea offered yet.

Resources & links

- ADEK K-12 AI Fluency Framework - The Phase 3 outcomes this unit is built on; teacher reference. (https://www.adek.gov.ae)
- Class responsible-use charter template - Printable poster for co-constructing safe-use norms. (teacher-prepared handout)

Week 1 - Lab | Rule vs Learning - Hands-On Exploration (Lab)

Strand: Integration | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Lab)

Objectives & success criteria

- Explore a rule-based demo and a simple learning demo and log the differences you observe.
- Devise a test that could tell whether a system follows rules or learns.
- Predict what would break each system and check your prediction.

Student-facing success criteria:

- I can run the rule-card machine and the learn-from-examples game and record what each does.
- I can log at least three differences I observed between them.
- I can design a fair test that would reveal whether a system follows rules or learns.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In this week's Core lesson you defined rule-based and learning systems and sorted examples. Now you investigate them hands-on so the difference becomes something you can see and test.

Activate: Recap the two definitions in ten seconds using the gear and seed icons.

Teach: Explain today's two stations: a rule-card machine that always follows the same steps, and a learn-from-examples card game where the group builds a pattern from cards.

Apply: Groups run both stations and log differences on the comparison sheet.

Check: Each group shares one difference and one prediction of what would break each system.

Critical & analytical thinking focus

Investigate & test - compare system behaviour (Analyse/Evaluate | DOK 2-3 | SOLO: Multistructural to Relational)

The technical concept (teacher background)

When you watch a rule-based system, its behaviour is completely predictable, because it always follows the same written steps. If you give it something its rules do not cover, it either stops or gives a wrong result. A learning system behaves differently - it was shown examples and built a model, so it can handle cases it has not seen exactly before, as long as they resemble its examples. But it also fails in a different way: if you show it something very unlike its training examples, it guesses badly. A good test to tell the two apart is to give the system a brand-new, unusual input and watch how it responds. Today's stations let you feel this difference without any devices. The rule-card machine shows fixed, repeatable behaviour; the card game shows a pattern being learned from examples the group chooses.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Set up stations 6 min	Explain the rule-card machine (a fixed if-then card stack) and the learn-from-examples game; assign groups and hand out the comparison log.	Read your role card and get your station materials ready.	• Which station will always do the same thing? • Which one might change based on the cards we pick?
Run the rule-card machine 10 min	Guide groups to feed inputs through the fixed if-then cards and record outputs.	Feed several inputs, including one strange one, and log exactly what happens each time.	• Did it ever change its behaviour? • What happened with the strange input?
Play the learn-from-examples game 12 min	Run the card game: the group is shown labelled example cards, then must classify new cards using only the pattern they saw.	Look at the example cards, build a pattern as a group, then classify the new cards and log your results.	• How did more examples change your guesses? • When did the group guess wrongly, and why?
Devise a test 10 min	Challenge each group to design one test that would reveal whether an unknown system follows rules or learns.	Write a test: what input would you give, and what result would show 'rules'	• What kind of input separates the two? • How would the results look

		versus 'learns'?	different?
Share and predict 7 min	Collect one difference and one 'what would break it' prediction from each group; add strong ideas to the question wall.	Share your best difference and predict what would break each system.	• Which system was more predictable? • Which one adapts, and what is its weakness?

Anticipated misconceptions

Misconception	How to address it
The learn-from-examples game is really just following hidden rules.	Point out that the group changed its pattern as new example cards appeared - a rule stack never changes. The pattern came from the examples, not from a fixed step list.
More example cards will always make the guesses better.	Show a case where adding a few odd or wrong cards made guesses worse. Quality and fairness of examples matter, not just how many there are.
A confident, fast guess in the game must be correct.	Reveal a confidently wrong guess. Speed and confidence do not equal accuracy, which is why we test on new cards and check results.

Hands-on activity guide

Overview: A two-station unplugged investigation. Station A is a rule-card machine (fixed if-then cards) that always behaves the same way. Station B is a learn-from-examples card game where groups build a pattern from labelled non-personal cards (shapes, objects, drawings) and classify new ones. Students log differences, then design a test to tell rules from learning.

Grouping / timing: Groups of three with rotating roles: feeder, recorder, tester.

Tool: Printed rule-card stack, labelled example cards (shapes/objects/drawings - no faces, names or voices), and the comparison log. - Fully unplugged and account-free. No student devices; no personal data is ever used.

Facilitation steps:

1. Assign roles: feeder, recorder, tester.
2. At Station A, feed five inputs through the rule cards, including one unusual input, and log each output.
3. Note whether Station A ever changed its behaviour.
4. At Station B, study the labelled example cards and agree a pattern as a group.
5. Classify five new cards using only the pattern you built; log your guesses and the correct answers.
6. Record at least three differences between the two stations.
7. Design one test that would reveal whether an unknown system follows rules or learns.
8. Predict what input would break each station and note it.

Unplugged fallback: If printed cards are unavailable, run both stations verbally: the teacher reads fixed if-then rules for Station A, and describes example items aloud for Station B while students classify by ear.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Checks whether students can observe and articulate the rule-based vs learning distinction through investigation

• Check: Comparison log with at least three logged differences and one devised test

• Success criteria: Logs at least three real differences and writes a test that would actually separate rules from learning.

• Evidence: Completed comparison log and the group's devised test.

• Feedback / adjustment: If tests do not separate the two, model one strong test (give an unseen, unusual input) and have groups revise.

Differentiation

Representation: Offer the log as a table with icons; provide a worked row as a model.

Action & expression: Roles let students contribute by feeding cards, recording, or testing - writing is not the only route.

Engagement: Use UAE-themed non-personal cards (dates, camels drawn as line-art, geometric Islamic patterns) for the learning game.

People of Determination: Give large-print cards, a card holder, and a partner scribe; allow the tester role to respond by pointing.

EAL: Provide sentence frames: 'Station A always ___', 'Station B changed when ___'; bilingual labels on cards.

Stretch / next tier: Ask groups to add a deliberately unfair set of example cards and explain how it changed the learned pattern.

Answer key / model responses

Task / item	Model response	Marking note
Log a difference between the stations.	Station A always gave the same output; Station B's guesses changed as we saw more example cards.	Any accurate observed difference is valid.
What happened with the unusual input at Station A?	It had no rule for it, so it stopped or gave a wrong answer.	Reinforces that rule-based systems fail on uncovered cases.
Design a test to tell rules from learning.	Give the system a brand-new, unusual input; if it always reacts the same way it is rule-based, if it adapts using patterns it likely learned.	Accept any test using a novel input to reveal adaptation.
Predict what would break Station B.	Showing it a card very unlike any example - it would guess badly.	Links to data quality and coverage.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Observation log	Logs at least three accurate differences with detail.	Logs one or two differences.	No clear differences logged yet.
Devising a test	Designs a test that would truly separate rules from learning.	Designs a test that partly separates them.	No workable test yet.
Prediction	Predicts what breaks each system with a sound reason.	Predicts for one system.	No reasoned prediction yet.

Resources & links

- Rule-card machine pack - Printable if-then cards for Station A. (teacher-prepared handout)
- Learn-from-examples card set - Labelled non-personal cards (shapes, objects, patterns) for Station B. (teacher-prepared handout)

Week 2 - Core | Rules vs Learning - How Each System Works

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G6·E | 1 x 45-minute period (Core)

Learning outcome (framework): Recognises the difference between rule-based and learning systems.

Objectives & success criteria

- Distinguish a rule-based system from a learning system using two clear features.
- Give one everyday example of each and explain the difference.
- Use the terms rule-based, learning, data and model accurately.

Student-facing success criteria:

- I can name two features that separate rule-based and learning systems.
- I can give one real example of each and explain why it fits.
- I can use the words rule-based, learning, data and model correctly.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Last week you launched the unit and investigated a rule-card machine and a learning game. Now you sharpen the distinction into clear, named features you can use to classify any system.

Activate: Flash four quick systems on the board; thumbs up for 'learns', thumbs down for 'rules'.

Teach: Build a two-column feature chart together: where rules come from, and how each system handles a new case.

Apply: Pairs complete the feature chart for two everyday UAE systems and name the distinguishing feature.

Check: Cold-call pairs to justify one classification using a feature, not a surface trait.

Critical & analytical thinking focus

Analyse & classify - contrast system types (Understand/Analyse | DOK 2 | SOLO: Multistructural)

The technical concept (teacher background)

A rule-based system and a learning system can do the same job yet work in completely different ways. In a rule-based system, a person decides the logic in advance and writes it as fixed steps; the data flows through those steps unchanged. In a learning system, a person does not write the logic - they collect data, label it, and let the system build a model that captures the pattern. Two features separate them cleanly. First, where the logic comes from: written by a person versus learned from data. Second, how they handle something new: a rule-based system can only handle cases its rules cover, while a learning system can generalise to new cases that resemble its examples. The model is the heart of a learning system - it is the pattern learned from the data, and it is only as good as the examples it was trained on. Knowing these features lets you classify almost any system you meet.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Warm-up sort 7 min	Flash four systems; students vote rules or learns and give a one-word reason.	Vote and share a quick reason for two of them.	• What made you vote that way? • Was it a surface clue or a real feature?
Build the feature chart 12 min	Co-construct a two-column chart: source of logic, handling of new cases, need for data; model the language rule-based, learning, data, model.	Copy the feature chart and add one example word to each row.	• Who writes the logic in each column? • Which column needs lots of data?
Classify everyday systems 12 min	Give pairs two UAE examples (a metro ticket gate; a voice assistant that improves with use) to classify using features.	Classify each example and name the feature that decides it.	• Which feature is the strongest evidence here? • Could a person write every rule for this?
Precision with words	Run a quick 'use the word correctly' round for data and model; correct loose usage.	Complete two sentences using data and model	• Is the model the same as the data?

8 min		accurately.	• What is the model built from?
Exit ticket 6 min	Issue the exit ticket: classify two systems and give the distinguishing feature.	Complete the exit ticket independently.	• What single feature best proves your answer? • Did you avoid a surface trait?

Anticipated misconceptions

Misconception	How to address it
You can tell a learning system just because it is on a phone or looks modern.	That is a surface trait. A calculator app is modern but rule-based. Judge by the features: where the logic comes from and how it handles new cases.
The data and the model are the same thing.	The data is the examples shown to the system; the model is the pattern it builds from that data. Different data can build a different model.
A learning system understands the meaning of what it classifies.	It matches patterns in data, it does not understand meaning. It can classify a cat photo without knowing what a cat is.

Hands-on activity guide

Overview: A structured classification activity: students build a feature chart, then use it to classify everyday UAE systems, defending each choice with a feature rather than a surface trait. Unplugged; any AI shown is a teacher board demo.

Grouping / timing: Pairs, with whole-class charting.

Tool: Feature-chart handout and everyday-system cards. - Account-free for students; teacher-only board demo for any live tool.

Facilitation steps:

1. Copy the two-column feature chart.
2. Add one example to each row with your partner.
3. Classify two everyday systems using the chart.
4. For each, underline the feature that decided your answer.
5. Write two sentences using data and model correctly.

Unplugged fallback: Runs fully unplugged using only the printed chart and cards; no demo required.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'recognises the difference between rule-based and learning systems'

• Check: Exit ticket: classify two systems and give the feature that distinguishes them

• Success criteria: Correctly classifies both and names a distinguishing feature (not a surface trait).

• Evidence: Exit slips sorted secure / partial / not-yet.

• Feedback / adjustment: Re-teach with a two-column feature chart if under 70 percent distinguish correctly; Primer next lesson.

Differentiation

Representation: Provide the feature chart pre-filled with icons for lower-literacy readers.

Action & expression: Let students classify by placing a token in a column or by writing.

Engagement: Use systems students meet daily in the UAE (metro gate, weather app, voice assistant).

People of Determination: Offer a two-choice token board and reduce the exit ticket to one system with support; allow verbal answers.

EAL: Bilingual glossary; sentence frame 'This is a ___ system because the logic is ___'.

Stretch / next tier: Ask students to find a system that is hard to classify and argue both sides using features.

Answer key / model responses

Task / item	Model response	Marking note
Classify a metro ticket gate.	Rule-based - a person set fixed rules (valid card opens the gate).	Distinguishing feature: logic written by a person.
Classify a voice assistant that improves with use.	Learning - it builds a model from lots of voice and text data and handles new phrases.	Distinguishing feature: handles new cases by generalising.
Use the word model in a sentence.	Example: 'The model is the pattern the system learned from many example photos.'	Check model is not confused with data.
Name two features that separate the systems.	Where the logic comes from, and how each handles a new case.	Both features must be non-surface.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Distinguishing features	Names two clear, non-surface features that separate the systems.	Names one feature or mixes in a surface trait.	No valid feature yet.
Classifying with justification	Classifies both examples correctly with a feature reason.	Classifies one correctly.	Cannot yet classify with a reason.
Using vocabulary	Uses data and model accurately and distinctly.	Uses one term accurately.	Terms confused or unused.

Resources & links

- Feature-chart handout - Two-column chart comparing rule-based and learning systems. (teacher-prepared handout)
- ADEK K-12 AI Fluency Framework - Outcome reference for AI Conceptual Understanding. (https://www.adek.gov.ae)

Week 2 - Lab | Train a Simple Model (Lab)

Strand: AI Conceptual Understanding | Framework ref(s): AIF·CU·G6·E | 1 x 45-minute period (Lab)

Learning outcome (framework): Recognises the difference between rule-based and learning systems.

Objectives & success criteria

- Train a simple classifier on non-personal examples (objects/drawings/sounds) and test it on new examples.
- Explain how the training examples shaped what the model learned.
- Run a basic risk and data-safety check on the training set.

Student-facing success criteria:

- I can sort examples into training and testing sets and train a simple classifier.
- I can test the model on new examples and record how it did.
- I can explain how my training examples changed what the model learned, and name one data-safety rule.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In this week's Core lesson you learned that a learning system builds a model from data. Now you build one yourself - either with a teachable image classifier on non-personal objects or with an unplugged paper-classifier - and see how the training examples shape it.

Activate: Show the training-to-testing flow diagram and recap: examples in, model built, new examples tested.

Teach: Explain the two golden rules: keep training and testing examples separate, and never use personal data (no faces, voices or names).

Apply: Groups build a training set of non-personal examples, train, then test on held-back examples and log results.

Check: Each group reports one result and one change that would improve the model.

Critical & analytical thinking focus

Build & investigate - train and test a model (Apply/Analyse/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Training a model means showing a learning system many labelled examples so it can find the pattern that separates the categories. We split our examples into two sets on purpose. The training set is what the model learns from. The testing set is examples the model has never seen, which we use to check whether it really learned the pattern or just memorised. If a model does well on training examples but poorly on new ones, it did not truly learn - this is why testing on fresh data matters. The examples you choose shape the model directly: if all your 'apple' pictures are red, the model may fail on a green apple, because it learned 'red' instead of 'apple'. That is why data quality, variety and fairness matter as much as quantity. A basic risk check asks two questions: are the examples safe and non-personal, and are they varied enough to be fair?

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Set the golden rules 6 min	Introduce the training/testing split and the data-safety rule (objects, drawings, shapes or sounds only - never faces, voices or names); show the flow diagram.	Note the two golden rules on your log.	• Why hold back some examples for testing? • Why never use a real face or name?
Build the training set 10 min	Guide groups to collect and label 8-10 non-personal examples per category and set aside 3 for testing.	Collect and label your training examples; keep your testing examples separate and hidden.	• Are your examples varied enough? • Did any personal data sneak in?
Train the model 8 min	Run the teachable classifier on the board (teacher-operated) with each group's set, or have groups build the paper-classifier pattern.	Watch or build your model from the training examples only.	• What pattern is the model picking up? • Did you train on testing examples by mistake?

Test on new examples 12 min	Feed each group's held-back testing examples and record correct/incorrect; discuss surprises.	Predict, then test each held-back example and log whether the model got it right.	• Did it work on examples it had never seen? • Which examples fooled it, and why?
Risk and improve 9 min	Lead a basic risk check and ask each group for one improvement to the training data.	Complete the data-safety and risk check; write one way to improve your training set.	• Was your data safe and fair? • What example would make the model stronger?

Anticipated misconceptions

Misconception	How to address it
Testing the model on the same examples it trained on proves it works.	That only shows it memorised. Real proof comes from testing on new, held-back examples the model has never seen.
Adding more examples always makes the model better.	Only if they are varied and fair. Ten red apples will not teach 'apple' - they teach 'red'. Variety and quality matter, not just quantity.
The model understands what an apple is.	It only found a pattern in pixels or sounds. It has no understanding, which is why an unusual example can fool it.

Hands-on activity guide

Overview: A substantial build-and-test lab. Groups create a training set of non-personal examples (objects, simple drawings, shapes or sounds), the teacher trains a school-approved child-safe teachable classifier on the board (or groups build an unplugged paper-classifier), then everyone tests the model on held-back examples and runs a data-safety and risk check. The training-to-testing flow and the data-safety rule are central.

Grouping / timing: Groups of three: data collector, labeller, tester.

Tool: A school-approved child-safe teachable image/sound classifier, operated by the teacher only, on non-personal data; OR an unplugged paper-classifier kit. - Account-free for students. If a digital classifier is used, only the teacher operates it, only on non-personal objects/drawings/shapes/sounds - never faces, voices or names.

Facilitation steps:

1. Choose two categories of non-personal items (for example, dates versus dried lemons, or circles versus triangles).
2. Collect and label 8-10 clear training examples per category.
3. Set aside 3 testing examples per category and keep them hidden.
4. Train the model (teacher-operated classifier or your paper-classifier) using training examples only.
5. Predict how each testing example will be classified.
6. Test each held-back example and log correct or incorrect.
7. Explain how your training examples shaped what the model learned.
8. Run the data-safety and risk check, then write one improvement to your training set.

Unplugged fallback: Unplugged paper-classifier: groups sort labelled cards into a decision chart built only from training cards, then classify held-back testing cards by hand - identical learning, no devices.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Deepens 'recognises the difference between rule-based and learning systems' by building a learning system and seeing how data shapes it

• Check: Training-and-testing log with results and an explanation of how the data shaped the model

• Success criteria: Trains and tests a model on separate example sets and explains how the training examples changed what it learned.

• Evidence: Completed training/testing log plus the data-safety and risk check.

• Feedback / adjustment: If groups train and test on the same examples, re-model the split and have them re-run on held-back data.

Differentiation

Representation: Provide the flow diagram and a worked example of a labelled training set.

Action & expression: Roles allow contribution through collecting, labelling or testing; verbal logging accepted.

Engagement: Use appealing UAE non-personal items (dates, geometric patterns, drawn camels, everyday classroom objects).

People of Determination: Provide pre-labelled example cards, large images, and a partner scribe; allow the tester to signal correct/incorrect by card.

EAL: Bilingual labels on categories; sentence frame 'The model learned ___ because my examples were ___'.

Stretch / next tier: Ask groups to design a 'trick' testing example and predict whether the model will fail and why.

Answer key / model responses

Task / item	Model response	Marking note
Why keep testing examples separate?	So we can check the model on things it never saw and know it truly learned, not memorised.	Core training/testing idea.
Explain how your training examples shaped the model.	Example: all our circles were large, so the model missed a small circle - it learned 'large' as well as 'round'.	Reward any accurate cause-and-effect from data to model.
Run the data-safety check.	All examples were objects or drawings, no faces, voices or names - safe.	Must confirm non-personal data.
Suggest one improvement.	Add varied examples (different sizes, colours, angles) so the model learns the real category.	Links quality and variety to accuracy.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Training and testing	Keeps sets separate, trains, and tests on new examples correctly.	Trains and tests but blurs the two sets.	Does not yet separate training and testing.
Explaining data effect	Clearly explains how the examples shaped the model.	Partial explanation.	No link between data and model yet.
Data safety and risk	Completes the safety check and names a real risk or improvement.	Partial safety check.	Safety check not completed.

Resources & links

- School-approved teachable classifier - Child-safe image/sound trainer, teacher-operated on non-personal data only. (school-approved tool (teacher account))
- Paper-classifier kit - Unplugged decision-chart and labelled non-personal cards. (teacher-prepared handout)

Week 3 - Core | Structured Prompts to Complete a Task

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G6·E | 1 x 45-minute period (Core)

Learning outcome (framework): Uses AI tools with structured prompts to complete tasks.

Objectives & success criteria

- Use a structured prompt to complete a defined task with an AI tool.
- Describe the input structure used and whether the task was completed.
- Adjust the prompt once to complete the task more fully.

Student-facing success criteria:

- I can write a structured prompt with a clear goal, context and format.
- I can say whether the task was completed and why.
- I can improve my prompt once to complete the task more fully.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You now know the difference between rule-based and learning systems. Many AI tools are learning systems, so the way you word your request - the prompt - changes what you get back. Today you learn to structure a prompt to complete a real task.

Activate: Show two prompts for the same task - a vague one and a structured one - and the two curated outputs they produced.

Teach: Introduce a simple prompt structure: Goal, Context, Format. Model completing one class task with it.

Apply: Pairs write a structured prompt for a defined task; the teacher runs it on the approved tool and shows the output.

Check: Pairs judge whether the task was completed and write one adjustment.

Critical & analytical thinking focus

Apply & evaluate - structured prompting to a goal (Apply/Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

A prompt is the instruction you give an AI tool, and because many AI tools are learning systems, the wording strongly shapes the result. A structured prompt states three things clearly: the goal (what you want done), the context (the details the tool needs), and the format (how the answer should look). A vague prompt leaves the tool guessing, so it often returns something off-target or incomplete. A structured prompt narrows the possibilities and makes the task far more likely to be completed. Prompting is not a one-shot skill - you evaluate the output against the task, then adjust the prompt to close the gap. This is the start of designing AI solutions: define the task, structure the input, check the output, and improve. Because the tool can produce confident but wrong answers, you always verify the result and never include personal data in the prompt.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Two prompts, two outputs 8 min	Show a vague prompt and a structured prompt for the same task with their curated outputs; ask which completed the task.	Compare the two outputs and note why one worked better.	• What did the vague prompt leave out? • Which output actually finished the task?
Teach Goal-Context-Format 10 min	Introduce the three-part structure; model writing one prompt for a class task (for example, summarise a short paragraph about the UAE for a Grade 4 reader in three bullet points).	Copy the structure and label the three parts in the model prompt.	• What is the goal here? • What context does the tool need? • What format did we ask for?
Write a structured prompt 12 min	Give pairs a defined task; circulate as they draft a Goal-Context-Format prompt; run selected prompts on the approved tool at the board.	Write a structured prompt for your task using all three parts.	• Have you stated the goal clearly? • Is the format specific?

Evaluate and adjust 9 min	Display outputs; guide pairs to judge task completion and make one improving adjustment; rerun one.	Decide if the task was completed, then adjust your prompt once to complete it more fully.	• What part of the task is missing? • Which single change would fix it?
Verify and log 6 min	Model a quick verification and risk check; students complete the task-completion log.	Verify the output against the task and finish your completion log.	• Is the output actually correct? • Did any personal data go into the prompt? It should not.

Anticipated misconceptions

Misconception	How to address it
A longer prompt is automatically a better prompt.	Structure beats length. A short prompt with a clear goal, context and format often beats a long, rambling one. Add detail only where it helps the task.
If the AI gives a confident, polished answer, the task is complete and correct.	A confident output can still be wrong or incomplete. Always compare it against the task and verify before trusting it.
You should put lots of personal details in a prompt so the AI knows who you are.	Never put personal or identifying data into a prompt. Give only the task details the tool needs; keep names, photos and private facts out.

Hands-on activity guide

Overview: A guided structured-prompting session. Pairs write Goal-Context-Format prompts for a defined, non-personal task; the teacher runs approved prompts on a school-approved child-safe tool and displays outputs. Pairs evaluate completion, make one adjustment, verify, and run a risk check.

Grouping / timing: Pairs, with teacher-mediated tool use at the board.

Tool: A school-approved child-safe AI tool, operated by the teacher only; students write prompts on paper first. - Account-free for students. All tool use is teacher-mediated on the board; no student logins; no personal data in any prompt.

Facilitation steps:

1. Read the defined task and underline what 'complete' means.
2. Write a structured prompt with Goal, Context and Format.
3. Hand your prompt to the teacher to run on the approved tool.
4. Judge whether the output completed the task.
5. Make one adjustment to your prompt to complete it more fully.
6. Verify the output against the task and complete a quick risk check.

Unplugged fallback: Use simulated-AI cards: pre-printed outputs for a vague prompt and a structured prompt so pairs practise structuring, evaluating and adjusting with no live tool.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'uses AI tools with structured prompts to complete tasks'

• Check: Task-completion log: structured prompt, output, 'task complete?' + one adjustment

• Success criteria: Completes the task using a structured prompt and records one adjustment that improved completion.

• Evidence: Task-completion log.

• Feedback / adjustment: Provide a stronger prompt structure for incomplete tasks; extend to multi-part prompts. Tool use is teacher-mediated and safe.

Differentiation

Representation: Provide a Goal-Context-Format prompt frame with an example filled in.

Action & expression: Students can dictate their prompt to a scribe or write it; evaluation can be spoken.

Engagement: Use tasks rooted in UAE topics students care about (a short summary about Sheikh Zayed's tree-planting, a packing list for a desert trip).

People of Determination: Offer a cloze prompt frame and reduce to completing one part at a time; allow verbal responses.

EAL: Bilingual labels for Goal, Context, Format; sentence frames for the evaluation.

Stretch / next tier: Ask students to add a fourth element (Audience) and show how it improved the output.

Answer key / model responses

Task / item	Model response	Marking note
Identify the three parts of a structured prompt.	Goal (what to do), Context (details the tool needs), Format (how the answer should look).	All three must be present.
Say whether a task was completed.	Example: 'Not fully - it gave four bullets but I asked for three, and one was off-topic.'	Look for judgement against the task, not just 'it looks good'.
Adjust the prompt once.	Example: add 'Give exactly three bullet points, each under 12 words.'	One clear change that improves completion.
Run a risk check.	No personal data was included; I verified the facts against the source before trusting them.	Both privacy and verification should appear.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Structured prompt	Writes a prompt with a clear goal, context and format.	Includes two of the three parts.	Prompt is vague or unstructured.
Evaluating completion	Judges completion accurately against the task.	Judges partly or vaguely.	No clear judgement yet.
Adjusting	Makes one adjustment that improves completion.	Makes a change with little effect.	No useful adjustment yet.

Resources & links

- Goal-Context-Format prompt frame - Printable scaffold for structuring prompts. (teacher-prepared handout)
- School-approved child-safe AI tool - Teacher-operated tool for board demonstrations. (school-approved tool (teacher account))

Week 3 - Lab | Prompt-to-Task Workshop (Lab)

Strand: AI Solution Design & Development | Framework ref(s): AIF·SD·G6·E | 1 x 45-minute period (Lab)

Learning outcome (framework): Uses AI tools with structured prompts to complete tasks.

Objectives & success criteria

- Complete a second, richer task and compare which prompt structure worked better.
- Design a reusable prompt structure for a class of tasks.
- Evaluate the outputs for accuracy and run a basic risk check.

Student-facing success criteria:

- I can complete a richer task with a structured prompt and compare two structures.
- I can design a reusable prompt template for similar tasks.
- I can check an output for accuracy and complete a risk check.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In this week's Core lesson you learned the Goal-Context-Format structure and completed one task. Now you tackle a richer task, compare prompt structures, and design a reusable template - the first step of the signature activity in later weeks.

Activate: Recap Goal-Context-Format and show two structured prompts that completed a task to different degrees.

Teach: Introduce comparing structures and designing a reusable template for a class of tasks (for example, 'summarise any short text for a younger reader').

Apply: Groups complete a richer task with two prompt structures, compare completion and accuracy, then draft a reusable template.

Check: Each group shares its template and one accuracy check it ran.

Critical & analytical thinking focus

Design & evaluate - build a reusable prompt structure (Create/Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

Once you can structure a single prompt, the next design step is to compare structures and reuse what works. A richer task - one with several requirements - shows the difference between prompt structures clearly, because a weak structure leaves parts of the task unfinished. By running two structures on the same task and comparing completion and accuracy, you gather evidence about which design is better, rather than guessing. From that evidence you can build a reusable template: a prompt skeleton with labelled slots that works for a whole class of similar tasks. This is real solution design - you are not just wording one request, you are building a tool others can reuse. Because outputs can be confidently wrong, every reusable template includes a verification step and a risk check, and no template ever asks for personal data. This workshop is the foundation for the signature activity, where you will evaluate a prompt-driven solution for accuracy and risk.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Recap and set the task 6 min	Recap Goal-Context-Format; present a richer task with three requirements and explain the compare-and-template goal.	Read the richer task and list its three requirements.	• What are the separate requirements here? • What would 'fully complete' look like?
Run two structures 13 min	Have groups draft two structured prompts (for example, one with format first, one with context first); run selected prompts on the approved tool and display outputs.	Draft two structured prompts for the same task and predict which will complete it better.	• Which structure covers all three requirements? • Where might each fall short?
Compare completion and accuracy	Guide groups to score each output for completion and verify accuracy against a source.	Compare the two outputs, mark which requirements were met, and check the	• Which structure completed more of the task? • Was every claim in the output

10 min		facts.	accurate?
Design a reusable template 10 min	Support groups to turn the winning structure into a labelled reusable template with a verify-and-risk step built in.	Write a reusable prompt template with labelled slots plus a verification and risk check.	• Which slots stay the same for every task? • Where does the verify step belong?
Share and risk check 6 min	Collect templates; run a quick class risk check on personal-data and accuracy.	Share your template and complete the risk check.	• Could this template leak personal data? It must not. • How does it force a check on accuracy?

Anticipated misconceptions

Misconception	How to address it
Once a template completes one task, it will work perfectly for every task without checking.	A template is a strong starting point, not a guarantee. Each new task still needs the output verified for completion and accuracy.
The structure that gives the longest answer is the best structure.	Judge by how fully and accurately it completes the task's requirements, not by length. A shorter, complete answer wins.
Accuracy checking is optional if the answer sounds right.	Sounding right is not being right. Every template must include a verification step against a reliable source.

Hands-on activity guide

Overview: A design workshop and the on-ramp to the signature activity. Groups complete a richer, multi-requirement task using two prompt structures, compare completion and accuracy, then design a reusable prompt template with a built-in verification and risk check. Tool use is teacher-mediated; an unplugged fallback uses prompt cards and curated outputs.

Grouping / timing: Groups of three: prompt writer, evaluator, template designer.

Tool: A school-approved child-safe AI tool operated by the teacher; prompt cards and curated outputs for the fallback. - Account-free for students. All live tool use is teacher-mediated on the board; no personal data in any prompt.

Facilitation steps:

1. List the three requirements of the richer task.
2. Draft two structured prompts that order Goal, Context and Format differently.
3. Predict which structure will complete the task more fully.
4. Have the teacher run both prompts; record the outputs.
5. Mark which requirements each output met and verify the facts.
6. Choose the stronger structure and turn it into a reusable template with labelled slots.
7. Build a verification step and a risk check into the template.
8. Test your template mentally on a second, similar task.

Unplugged fallback: Unplugged prompt-card workshop: groups arrange Goal-Context-Format cards into two structures and match them to curated printed outputs, then design their template on paper - no live tool needed.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Extends 'uses AI tools with structured prompts to complete tasks' toward reusable, evaluated solutions

• Check: A reusable prompt template plus a comparison of two structures with accuracy checks

• Success criteria: Compares two structures on completion and accuracy and designs a reusable template with a verification step.

• Evidence: Comparison notes and the reusable template with its risk check.

• Feedback / adjustment: If templates omit verification, model a simple check-against-source step and have groups add it.

Differentiation

Representation: Provide a template skeleton with labelled empty slots and a worked example.

Action & expression: Roles let students contribute by writing prompts, evaluating, or designing the template; verbal contribution accepted.

Engagement: Use motivating UAE tasks (a visitor guide to a local souq, a short recycling notice for the class).

People of Determination: Offer a partly built template and a checklist for the comparison; allow the evaluator to respond by ticking a chart.

EAL: Bilingual slot labels; sentence frame 'Structure A completed ___ requirements; Structure B completed ___'.

Stretch / next tier: Ask groups to test their template on a genuinely different task type and report where it needed changing.

Answer key / model responses

Task / item	Model response	Marking note
List the requirements of the richer task.	Example for a souq visitor guide: three tips, each one sentence, friendly tone.	Requirements must be specific and checkable.
Compare the two structures.	Example: 'Structure A met all three requirements; Structure B missed the tone, so A was stronger.'	Comparison must be on completion or accuracy, not length.
Design a reusable template.	Goal: [task]; Context: [details, no personal data]; Format: [exact format]; then Verify: check facts against a source.	Template must include a verify/risk step.
Run the risk check.	No personal data requested; output verified against a reliable source before use.	Both privacy and accuracy must appear.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Comparing structures	Compares two structures on completion and accuracy with evidence.	Compares loosely or on one measure.	No real comparison yet.
Reusable template	Designs a clear reusable template with labelled slots.	Designs a partial template.	No workable template yet.
Accuracy and risk	Builds in a verification step and completes the risk check.	Includes one of verification or risk check.	Neither included yet.

Resources & links

- Reusable template skeleton - Printable prompt template with labelled slots and a verify step. (teacher-prepared handout)
- Prompt-card and curated-output pack - Unplugged fallback materials for the workshop. (teacher-prepared handout)

Week 4 - Core | Judging Outputs - Correct, Incorrect, Unclear

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G6·E | 1 x 45-minute period (Core)

Learning outcome (framework): Identifies when outputs are incorrect or unclear.

Objectives & success criteria

- Identify, in a set of outputs, which are incorrect or unclear and say why.
- Explain one signal that an output cannot be trusted as-is.
- Recommend one action to take when an output is unclear.

Student-facing success criteria:

- I can sort outputs into correct, incorrect, and unclear.
- I can give a reason for each judgement I make.
- I can name one action to take when an output cannot be trusted.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Grade 5 you learned that AI tools can be confidently wrong. In earlier Grade 6 weeks you saw how a system type and a prompt shape an output. Today you turn that into a habit: checking an output before you use it.

Activate: Show two answers to the same question - one right, one wrong but confident. Ask: which would you trust, and how could you tell?

Teach: Explain three verdicts an output can earn: correct (matches the task and a source), incorrect (clashes with what we can check), unclear (we cannot yet tell). Model checking one output aloud.

Apply: In pairs, students give a verdict to one board-displayed output and write the reason in six words or fewer.

Check: Cold-call three pairs: verdict plus reason. Correct any 'it just feels wrong' with a checkable reason.

Critical & analytical thinking focus

Evaluate - scrutinise outputs for correctness/clarity (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

An AI language model produces text by predicting likely word patterns, not by looking up guaranteed facts. This means an output can sound fluent and confident while still being wrong, a problem often called a hallucination. Because the model does not know whether it is right, the job of judging correctness falls to you, the human user. A useful routine is to give every output one of three verdicts: correct, incorrect, or unclear. Correct means it matches the task you set and agrees with a source you trust. Incorrect means it clashes with something you can check, such as a fact, a number, or the instruction you gave. Unclear means the output is vague, incomplete, or you simply cannot verify it yet, so it needs more checking before use. Learning to separate 'sounds right' from 'is right' is the core skill of critical evaluation, and it protects you from acting on a confident mistake.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Hook and activate 7 min	Display two confident answers to 'How many bones are in the human body?' - one correct (206), one wrong (198). Ask students to vote and justify.	Vote, then explain to a partner how they could find out which answer is right.	• Which answer do you trust more, and why? • Does sounding sure make an answer correct?
Teach the triage routine 10 min	Introduce the three verdicts (correct / incorrect / unclear). Model a think-aloud: read the task, read the output, check against a source, assign a verdict with a reason.	Copy the three-verdict routine into their notes and follow one modelled example.	• What is the difference between incorrect and unclear? • What could you check this against?
Guided practice 12 min	Show three curated outputs (a fact answer, a summary that adds a detail not in the source, a vague instruction). Guide	In pairs, assign verdicts and reasons to outputs 2 and 3 on a mini whiteboard.	• What signal here says 'do not trust as-is'? • Which single detail makes this

	the class to triage the first together.		incorrect?
Discuss signals and actions 9 min	Collect the signals students used (missing source, contradicts task, too vague, made-up detail). Link each unclear or incorrect verdict to an action: re-check, re-prompt, or ask a person.	Add one 'signal' and one matching 'action' to a class list.	• When an output is unclear, what is your next move? • Who could you ask to verify?
Exit check 7 min	Hand out the output-triage exit card: three short outputs to flag correct / incorrect / unclear with a reason each.	Complete the exit card independently.	• Can you give a checkable reason, not just a feeling?

Anticipated misconceptions

Misconception	How to address it
If an answer sounds confident and detailed, it must be correct.	Show a confident but wrong output. Fluent language is a writing style, not proof. Ask: what did we check it against?
Incorrect and unclear are the same thing.	Incorrect clashes with something we can check now; unclear means we cannot tell yet and need more information. Sort one example into each.
If I cannot verify an output, I should just use it anyway.	Unclear means pause and check, not proceed. Model choosing an action: re-prompt, find a source, or ask a person.

Hands-on activity guide

Overview: Short paper triage: students judge three curated outputs and label each with a reason, building the routine used fully in the Lab.

Grouping / timing: Pairs, then independent exit card.

Tool: Teacher board demo only; students use printed output cards (account-free). - No student accounts. Teacher displays curated outputs; students work on paper.

Facilitation steps:

1. Read the task each output was meant to answer.
2. Read the output and check it against the task and any given source.
3. Assign a verdict: correct, incorrect, or unclear.
4. Write a six-word reason for the verdict.
5. Name one action for any unclear or incorrect output.

Unplugged fallback: If no board is available, distribute the same outputs as printed cards with a reference sheet.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'identifies when outputs are incorrect or unclear'

• Check: Output-triage exit card: flag correct / incorrect / unclear with a reason each

• Success criteria: Correctly triages at least 2 of 3 outputs and gives a valid reason for each judgement.

• Evidence: Output-triage card.

• Feedback / adjustment: Model 'check against the task and a source' for over-trusting students; extend to subtle inaccuracies.

Differentiation

Representation: Colour-code the three verdicts; pair each with an icon (tick, cross, question).

Action & expression: Allow verbal verdicts recorded by a partner instead of written reasons.

Engagement: Use a 'trust detective' framing so students hunt for the giveaway signal.

People of Determination: Provide a printed verdict-and-reason frame with the routine steps numbered; allow extra time and a scribe.

EAL: Give a bilingual glossary (correct, incorrect, unclear, source, verify) and sentence stems.

Stretch / next tier: Ask stretch students to find the single subtle detail that flips an output from correct to incorrect.

Answer key / model responses

Task / item	Model response	Marking note
Triage: 'The capital of the UAE is Abu Dhabi.'	Correct - matches a trusted source and the task.	Reason must reference a check, not a feeling.
Triage: a summary that adds a prize not mentioned in the original text.	Incorrect - it states a detail that clashes with the source.	Added detail is a classic hallucination signal.
Triage: 'Do the thing with the numbers to get the result.'	Unclear - too vague to verify; needs a clearer prompt.	Action: re-prompt for specific steps.
Name one signal an output cannot be trusted as-is.	It contradicts the task, has no source, or contains a detail we cannot check.	Any one checkable signal is acceptable.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Assigns correct verdicts	Triages at least 2 of 3 outputs correctly.	Triages 1 of 3 correctly.	Verdicts do not match the outputs.
Gives checkable reasons	Each verdict has a reason tied to task or source.	Some reasons rely on feeling not evidence.	Reasons missing or unrelated.
Recommends an action	Names a sensible action for an unclear output.	Names a vague action.	No action given.

Resources & links

- Curated output-triage card set - Teacher-prepared correct/incorrect/unclear examples with sources. (teacher-prepared)
- ADEK K-12 AI Fluency Framework - Reference for the Critical Evaluation strand outcomes. (https://www.adek.gov.ae)

Week 4 - Lab | Output Triage Clinic (Lab)

Strand: Critical Evaluation & Informed Interaction | Framework ref(s): AIF·CE·G6·E | 1 x 45-minute period (Lab)

Learning outcome (framework): Identifies when outputs are incorrect or unclear.

Objectives & success criteria

- Apply a triage routine to a set of curated outputs and record correct/incorrect/unclear with reasons.
- Rank outputs from most to least trustworthy and defend the order.
- Choose a verification action for each untrustworthy output.

Student-facing success criteria:

- I can run the triage routine across a full set of outputs.
- I can rank outputs by trustworthiness and defend my order.
- I can choose a verification action for outputs I cannot trust.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you learned the three-verdict routine and practised it on three outputs. In the Lab you run a full clinic, triaging a larger set and ranking them.

Activate: Show a 'trust ladder' with two outputs already placed. Ask where a third belongs and why.

Teach: Explain the clinic stations and the record sheet columns; model triaging and placing one output on the ladder.

Apply: Groups triage one station output together as a warm-up.

Check: Each group reads out one verdict and its verification action.

Critical & analytical thinking focus

Evaluate - scrutinise outputs for correctness/clarity (Evaluate | DOK 2-3 | SOLO: Relational)

The technical concept (teacher background)

Verification is the practical side of critical evaluation: not just noticing an output might be wrong, but taking a concrete step to check it. Different outputs need different checks - a factual claim can be compared to a reliable source, a summary can be compared back to the original text, and a vague instruction may need a clearer prompt before it can be judged at all. Ranking outputs by trustworthiness forces you to weigh these signals against one another rather than treating every answer the same. In a clinic, you also practise a basic risk check: asking what could go wrong if someone used this output as-is. Because the model cannot self-verify, this human checking layer is where safety and accuracy actually come from. The routine you rehearse here becomes automatic and travels into every later task where you use an AI solution.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Set up the clinic 8 min	Explain stations, record sheet, and the trust ladder. Model triaging one output and placing it on the ladder with a spoken reason.	Form groups, collect record sheets, read the routine steps.	• What are the three verdicts? • What will you check each output against?
Triage stations 15 min	Rotate between stations of curated outputs (fact claim, summary with added detail, unclear instruction, near-correct number). Prompt groups to name a verification action for each.	Triage each station output, record verdict, reason, and a verification action.	• What single detail decides this verdict? • How would you verify it?
Rank on the trust ladder 10 min	Ask each group to order their outputs from most to least trustworthy and prepare a defence of the order.	Place outputs on the ladder and agree a group justification.	• Why is this one more trustworthy than that one? • What would move it up the ladder?
Share and risk	Two groups present their ladder; class adds a basic risk check: what could go	Present or critique one ranking and name one risk	• Who could be affected if this

check 7 min	wrong if the least trustworthy output were used?	of using an untrustworthy output.	were used as-is?
Reflect 5 min	Collect record sheets; give one clinic each a 'focus for next time'.	Write one verification action they will reuse.	• Which check was most useful today?

Anticipated misconceptions

Misconception	How to address it
Ranking is just about which answer I like best.	Ranking uses trustworthiness signals - sources, matches to task, checkability - not preference. Ask for the signal behind each placement.
Every output needs the same kind of check.	A fact needs a source; a summary needs the original text; a vague instruction needs a clearer prompt. Match the check to the output type.
An unclear output is useless.	Unclear means not yet judged; a verification action can often move it to correct or incorrect.

Hands-on activity guide

Overview: A rotating triage clinic where groups judge a curated set of outputs, choose a verification action for each, then rank all outputs on a trust ladder and defend the order.

Grouping / timing: Groups of 3-4 rotating through 4 stations.

Tool: Teacher-mediated board demo plus printed station cards; no student accounts. - Curated outputs prepared by the teacher and displayed or printed; students never enter data into a tool.

Facilitation steps:

1. At each station, read the task and the curated output.
2. Assign a verdict: correct, incorrect, or unclear.
3. Write a checkable reason for the verdict.
4. Choose a verification action (check a source, compare to original, re-prompt, ask a person).
5. After all stations, rank outputs on the trust ladder from most to least trustworthy.
6. Agree a group defence for the top and bottom placements.
7. Name one risk of using the least trustworthy output as-is.

Unplugged fallback: Unplugged: use printed output cards and a reference sheet; groups sort and rank on a paper ladder without any device.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'identifies when outputs are incorrect or unclear'

• Check: Completed clinic record sheet plus a defended trust-ladder ranking

• Success criteria: Triages the full set with valid reasons, chooses a verification action each, and defends a sensible ranking.

• Evidence: Clinic record sheet and trust ladder photo.

• Feedback / adjustment: For over-trusting groups, model matching a check to the output type; extend ranking to justify near-ties.

Differentiation

Representation: Provide a laminated routine card and colour-coded verdict tokens.

Action & expression: Allow groups to place physical tokens on the ladder instead of writing.

Engagement: Frame as a 'trust clinic' where each output is a patient needing a diagnosis.

People of Determination: Assign a clear role (reader, recorder, ranker) and give a numbered step card; permit extra time and a scribe.

EAL: Bilingual glossary and pre-labelled ladder rungs (most to least trustworthy) with icons.

Stretch / next tier: Require a written defence of one near-tie on the ladder and a proposed action to break the tie.

Answer key / model responses

Task / item	Model response	Marking note
Station A fact claim about a UAE landmark.	Correct if it matches a trusted source; verification action = check official site.	Reason must cite the source used.
Station B summary with an added award.	Incorrect - added detail not in the original; action = compare to source text.	Signal: information appears that was never in the source.
Station C vague instruction.	Unclear; action = re-prompt for specific steps.	Cannot be judged until clarified.
Rank the four outputs.	Most trustworthy = verified fact; least = unverified added-detail summary.	Any defensible order with signal-based reasons is acceptable.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Runs the triage routine	Triages the full set with valid, checkable reasons.	Triages most, some reasons weak.	Routine not applied consistently.
Chooses verification actions	Matches a suitable check to each output type.	Uses one generic check for all.	No verification action chosen.
Defends the ranking	Ranking justified with trustworthiness signals.	Ranking given with weak justification.	Ranking based on preference only.

Resources & links

- Triage clinic station pack - Four curated outputs with matched sources for verification. (teacher-prepared)
- Trust ladder template - Printable ladder for ranking outputs by trustworthiness. (teacher-prepared)

Week 5 - Core | Risks and Responsible Use

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G6·E | 1 x 45-minute period (Core)

Learning outcome (framework): Recognises basic risks (e.g., incorrect outputs, unsafe use).

Objectives & success criteria

- Identify at least two basic risks of using AI tools (e.g., incorrect output, unsafe or private data use).
- Explain why each risk matters and who it could affect.
- State one responsible-use rule that reduces a risk.

Student-facing success criteria:

- I can name two genuine risks of using an AI tool.
- I can explain who each risk could affect.
- I can state a responsible-use rule that reduces a risk.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Last week you learned to judge whether an output is correct, incorrect, or unclear. This week you zoom out: even a good-looking output carries risks, and responsible use means planning for them before they cause harm.

Activate: Show a short scenario where a student pastes a classmate's full name and address into an AI tool. Ask: what could go wrong here?

Teach: Introduce two risk families: accuracy risk (incorrect output used as-is) and safety/privacy risk (unsafe or private data use). Give a clear example of each.

Apply: Pairs match three scenarios to a risk family and name who is affected.

Check: Cold-call: one risk, who it affects, and a rule that reduces it.

Critical & analytical thinking focus

Evaluate - identify and reason about risk (Evaluate | DOK 2 | SOLO: Relational)

The technical concept (teacher background)

Using an AI tool responsibly means thinking about risk before something goes wrong. Two basic risk families matter most at this stage. The first is accuracy risk: because a model can produce confident but incorrect outputs, acting on one without checking can spread mistakes into your schoolwork or advice you give others. The second is safety and privacy risk: anything you type into a tool may be stored or used to train future systems, so entering personal or identifying data - names, addresses, photos, ID numbers - can expose you or someone else. Every risk has a who: the person affected, which may be you, a classmate, or your family. Responsible-use rules are simple, agreed habits that reduce a risk, such as 'always verify before you use' or 'never enter personal data'. Governance is just the grown-up word for having and following such rules together as a class, a school, or a country. Recognising risk early is what lets you enjoy the benefits of AI while keeping people safe.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Hook and activate 7 min	Present the 'pasted personal data' scenario and a 'used a wrong AI answer in homework' scenario. Ask what could go wrong in each.	Discuss with a partner and name one thing that could go wrong per scenario.	• What is the risk here? • Who could be harmed?
Teach the two risk families 10 min	Define accuracy risk and safety/privacy risk. Give a clear example of each and name the person affected.	Note the two families and one example each.	• Which family does each scenario belong to? • Why does this risk matter?
Guided practice - who is affected 12 min	Show three more scenarios. Guide the class to name the risk, the person affected, and a responsible-use rule.	In pairs, complete a risk-who-rule chain for two scenarios.	• Who is affected if nobody checks? • What rule would have prevented this?

Build the responsible-use rules 9 min	Collect student rules into a short class charter (e.g., verify before use, never enter personal data, tell an adult if unsure).	Contribute one rule and match it to a risk it reduces.	• Which risk does your rule reduce? • Is this rule realistic to follow?
Exit check 7 min	Hand out the risk-spotting exit card: name two risks plus one responsible-use rule.	Complete the exit card independently.	• Are your two risks genuine and different?

Anticipated misconceptions

Misconception	How to address it
AI tools are either totally safe or totally dangerous.	Risk is about specific situations, not the whole tool. Sort one safe use and one risky use of the same tool.
It is fine to type in anyone's personal details as long as it helps.	Personal data may be stored or used to train systems. Model replacing a real name with 'a student' in a prompt.
Saying 'it is bad' is enough to describe a risk.	A real risk names what could go wrong and who it affects. Rewrite 'it is bad' into a specific risk plus a person.

Hands-on activity guide

Overview: Scenario sorting: students match scenarios to a risk family, name who is affected, and attach a responsible-use rule, previewing the full audit in the Lab.

Grouping / timing: Pairs, then independent exit card.

Tool: Teacher-mediated board demo; printed scenario cards; account-free. - No student accounts; teacher displays or prints scenarios.

Facilitation steps:

1. Read the scenario card.
2. Decide the risk family: accuracy or safety/privacy.
3. Name who could be affected.
4. Attach one responsible-use rule that reduces the risk.

Unplugged fallback: Use printed scenario cards and a rules reference sheet with no device.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'recognises basic risks (incorrect outputs, unsafe use)'

• Check: Risk-spotting exit card: name two risks + one responsible-use rule

• Success criteria: Names two genuine risks, explains who is affected, and states a matching responsible-use rule.

• Evidence: Risk-spotting card.

• Feedback / adjustment: Redirect vague 'it is bad' answers to a specific risk + who it affects; extend to privacy risks.

Differentiation

Representation: Use two clearly labelled risk-family icons (a cross for accuracy, a lock for privacy).

Action & expression: Allow students to sort physical scenario cards instead of writing.

Engagement: Frame as a 'risk spotter' role protecting the class.

People of Determination: Provide a risk-who-rule sentence frame and a printed rules bank; allow extra time.

EAL: Bilingual glossary (risk, privacy, personal data, responsible use, affect) and stems.

Stretch / next tier: Ask students to find a scenario that involves both risk families at once and explain the overlap.

Answer key / model responses

Task / item	Model response	Marking note

Name two basic risks of using AI tools.	Incorrect output used as-is (accuracy) and entering personal data (safety/privacy).	The two risks must be genuinely different.
Who is affected if a wrong AI answer is used in homework?	The student using it, and anyone who trusts their work.	Naming a specific 'who' is required.
State a responsible-use rule that reduces the privacy risk.	Never enter personal or identifying data into an AI tool.	Rule must match the named risk.
Why does the accuracy risk matter?	A confident wrong answer can spread mistakes if used without checking.	Should link risk to a consequence.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Identifies genuine risks	Names two different, genuine risks.	Names one genuine risk.	Risks vague or not genuine.
Explains who is affected	Names a specific person or group per risk.	Names who for one risk only.	No 'who' given.
States a matching rule	Rule clearly reduces a named risk.	Rule loosely related.	No rule or unrelated rule.

Resources & links

- AI risk scenario cards - Teacher-prepared accuracy and privacy scenarios set in UAE school life. (teacher-prepared)
- ADEK responsible-use guidance - Reference for the Ethics & Governance strand and school rules. (https://www.adek.gov.ae)

Week 5 - Lab | Class AI Risk Audit (Lab)

Strand: Ethics & Governance | Framework ref(s): AIF·GE·G6·E | 1 x 45-minute period (Lab)

Learning outcome (framework): Recognises basic risks (e.g., incorrect outputs, unsafe use).

Objectives & success criteria

- Audit the class's own AI use against a responsible-use rule set and propose one improvement.
- Rank risks by likelihood and impact on a simple risk matrix and choose which to mitigate first.
- Write one responsible-use rule to close a gap found in the audit.

Student-facing success criteria:

- I can audit our AI use against a rule set and find a gap.
- I can place risks on a matrix by likelihood and impact.
- I can propose one rule to reduce the highest-priority risk.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you learned two risk families and how to name who is affected. In the Lab you audit your own class's AI use, rank the risks, and decide what to fix first.

Activate: Show an anonymised list of how the class has used AI demos so far. Ask: where might a risk be hiding?

Teach: Explain the audit checklist and the risk matrix; model auditing one use and placing its risk on the matrix.

Apply: Groups audit one class use together as a warm-up.

Check: Each group states one risk and where it sits on the matrix.

Critical & analytical thinking focus

Evaluate - identify and reason about risk (Evaluate | DOK 2 | SOLO: Relational)

The technical concept (teacher background)

Auditing means checking real practice against a set of rules to find gaps. In an AI risk audit you look at how AI has actually been used - what was typed in, what outputs were trusted, whether any personal data appeared - and compare it to your responsible-use rules. Not every risk deserves the same attention, so you weigh each one on two dimensions: likelihood (how often it could happen) and impact (how much harm it would cause). Plotting risks on a matrix helps you prioritise: a high-likelihood, high-impact risk should be mitigated before a rare, low-harm one. Mitigation means adding or strengthening a rule to reduce a risk. This is exactly how organisations and governments approach governance, just at a class scale. Because your audit uses only anonymised, non-identifying examples, it teaches the skill without creating the very privacy risk it studies.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Set up the audit 8 min	Present the anonymised class-use log and the audit checklist. Model auditing one entry and placing its risk on the matrix.	Form groups, read the checklist, note the two matrix axes.	• What are we comparing our use against? • What do likelihood and impact mean?
Run the audit 14 min	Groups work through the anonymised use log, flagging each entry against the rule set and identifying gaps.	Audit entries, record gaps found, and note the risk behind each gap.	• Which rule does this use break or stretch? • Who could be affected?
Plot the risk matrix 10 min	Guide groups to place their flagged risks on the likelihood-impact matrix and pick the highest priority.	Place each risk on the matrix and circle the one to mitigate first.	• Why is this risk higher priority than that one? • What makes it high impact?
Propose a rule 8 min	Each group proposes one responsible-use rule to close its top-priority gap; class votes to add the strongest to the charter.	Write and pitch one mitigation rule.	• Will this rule actually reduce the risk? • Is it realistic to follow?

Reflect 5 min	Collect audit sheets; give each group a 'focus for responsible use'.	Write one change they will make to their own AI use.	• What is your personal takeaway from the audit?

Anticipated misconceptions

Misconception	How to address it
An audit is about blaming whoever made a mistake.	The log is anonymised; audits fix systems, not people. Reframe a flagged entry as a gap to close, not a person to blame.
All risks should be fixed at once and equally.	Use the matrix to prioritise. Mitigate high-likelihood, high-impact risks first; model placing two risks and choosing.
A rule counts even if nobody could realistically follow it.	Test each rule for realism. Rewrite an impractical rule into one the class can actually keep.

Hands-on activity guide

Overview: A structured, teacher-mediated audit of the class's own AI use against a responsible-use rule set, followed by ranking the flagged risks on a likelihood-impact matrix and proposing one mitigation rule.

Grouping / timing: Groups of 3-4; whole-class charter vote at the end.

Tool: Unplugged and teacher-mediated: anonymised paper use-log and printed matrix; no student accounts. - Only anonymised, non-identifying examples are used; students never enter real data anywhere.

Facilitation steps:

1. Read the anonymised class AI-use log.
2. Check each entry against the responsible-use rule set.
3. Flag any gap and name the risk and who it affects.
4. Place each flagged risk on the likelihood-impact matrix.
5. Circle the highest-priority risk to mitigate first.
6. Write one responsible-use rule that closes that gap.
7. Pitch the rule for the class charter.

Unplugged fallback: Fully unplugged already; if the log is unavailable, use pre-written anonymised scenario cards instead.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Directly checks 'recognises basic risks (incorrect outputs, unsafe use)'

• Check: Completed audit sheet with a matrix-placed priority risk and a proposed mitigation rule

• Success criteria: Finds a genuine gap, places risks sensibly on the matrix, and proposes a realistic rule that reduces the top risk.

• Evidence: Audit sheet and risk matrix photo.

• Feedback / adjustment: For blame-focused groups, redirect to fixing the system; extend by asking for a second-priority risk and rule.

Differentiation

Representation: Provide the matrix pre-drawn with labelled axes and a worked example placed.

Action & expression: Allow risks to be placed as sticky notes rather than written into cells.

Engagement: Frame groups as a 'class safety board' advising the school.

People of Determination: Assign roles (reader, flagger, plotter) and give a numbered step card; allow extra time and a scribe.

EAL: Bilingual glossary (audit, likelihood, impact, mitigate, gap) and matrix axis labels with icons.

Stretch / next tier: Ask groups to justify a near-tie placement and propose how a rule would move the risk down the matrix.

Answer key / model responses

Task / item	Model response	Marking note

Audit an entry where a demo prompt included a student's name.	Gap: personal data entered; risk = privacy exposure; who = that student.	Should map to the 'never enter personal data' rule.
Place the privacy risk on the matrix.	High impact; likelihood depends on habits - often high without a rule.	Placement must be justified with a reason.
Choose which risk to mitigate first.	The high-likelihood, high-impact one (usually privacy).	Priority follows matrix position.
Propose a mitigation rule.	Replace any real name with 'a student' before using a prompt.	Rule must be realistic and reduce the named risk.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Finds a genuine gap	Identifies a real gap and its risk and who.	Identifies a gap but not the who.	No genuine gap found.
Uses the risk matrix	Places risks sensibly and picks a justified priority.	Places risks but priority weakly justified.	Matrix not used to prioritise.
Proposes a mitigation rule	Realistic rule that reduces the top risk.	Rule loosely related or hard to follow.	No workable rule proposed.

Resources & links

- Anonymised class AI-use log - Teacher-prepared, non-identifying record of how demos were used. (teacher-prepared)
- Likelihood-impact risk matrix template - Printable 2x2 matrix for ranking risks. (teacher-prepared)

Week 6 - Core | Bringing the Strands Together

Strand: Integration | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Core)

Objectives & success criteria

- Complete a guided challenge combining the concept, a prompted solution, an output check and a risk check.
- Explain how the system type and prompt shaped the output.
- Identify one responsible-use choice made.

Student-facing success criteria:

- I can complete a guided challenge that uses all four strands.
- I can explain how the system type and prompt shaped the output.
- I can name one responsible-use choice I made.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Across Term 1 you learned four strands: rule-based versus learning systems (concept), writing prompts (solution design), judging outputs (critical evaluation), and spotting risks (ethics and governance). Today you weave all four into one guided challenge.

Activate: Show four strand icons and ask students to match each to something they can now do.

Teach: Model the four-step chain on one mini task: name the system type, write a prompt, triage the output, run a risk check.

Apply: Pairs complete step one and two of a shared mini challenge on the board.

Check: Cold-call: how did the prompt change the output?

Critical & analytical thinking focus

Synthesise - connect all four strands (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Real AI fluency is not four separate skills but one connected process. It starts with understanding the system: a rule-based system follows fixed instructions, while a learning system finds patterns in data, and knowing which you face shapes what you can expect. Next comes solution design: a clear, specific prompt guides the system toward a useful output, while a vague prompt invites a vague or wrong one. Then comes critical evaluation: you triage the output as correct, incorrect, or unclear and verify it against a source. Finally comes ethics and governance: you run a basic risk check and make a responsible-use choice, such as keeping personal data out of the prompt. Each strand feeds the next - the system type shapes the prompt, the prompt shapes the output, the output demands a check, and the whole process demands responsibility. Seeing these links is what turns isolated lessons into judgement you can use on any new AI task.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Recall the four strands 7 min	Display the four strand icons and a one-line reminder of each. Ask students to connect them into a chain.	Order the four strands into a process chain with a partner.	• Which strand comes first, and why? • How does one strand feed the next?
Model the integrated chain 10 min	Take one guided challenge (e.g., 'summarise a short UAE history passage'). Model naming the system type, writing a prompt, triaging the output, and a risk check.	Follow the modelled chain and note each step.	• How did the prompt shape the output? • What risk did we need to check?
Guided challenge 13 min	Give pairs a new guided challenge with a curated output. Guide them through all four steps against the project success criteria.	Complete the four-step chain and record each result.	• Is your output correct, incorrect, or unclear? • What responsible-use choice did you make?
Connect and	Ask pairs to explain how the system type	Write a two-sentence	• Which strand was strongest in

explain 8 min	and prompt shaped the output, and name one responsible-use choice.	explanation linking system, prompt, and output.	your work? • Which needs more practice?
Exit and feedforward 7 min	Review the integration task against success criteria; give each student one 'focus for planning' tied to a weaker strand.	Record their focus strand for the signature solution.	• Which strand is your focus before the big task?

Anticipated misconceptions

Misconception	How to address it
The four strands are separate skills used one at a time.	They form a chain where each feeds the next. Trace the chain aloud on one task: system to prompt to output to risk.
Once the output looks good, the job is done.	A good-looking output still needs a triage and a risk check. Model adding both after a plausible output.
Responsible use is a separate add-on, not part of the task.	Show that the responsible-use choice (e.g., no personal data) is made during the prompt, not after.

Hands-on activity guide

Overview: A guided integration challenge where students run one task through all four strands and explain the links, previewing the signature solution.

Grouping / timing: Pairs, then independent explanation and exit.

Tool: Teacher-mediated board demo with a curated output; account-free. - No student accounts; teacher runs any demo and displays curated outputs.

Facilitation steps:

1. Name the system type behind the task.
2. Write or choose a clear prompt.
3. Triage the curated output as correct, incorrect, or unclear.
4. Run a basic risk check and note one responsible-use choice.
5. Explain how the system and prompt shaped the output.

Unplugged fallback: Use a fully unplugged version with printed task, prompt options, and a curated output card.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Integrates all Term-1 Emerging outcomes before the signature solution
• Check: Guided integration task reviewed against project success criteria; forward feedback
• Success criteria: Connects system+prompt to output, triages the output, and names a risk-reduction choice.
• Evidence: Integration task + feedback note.
• Feedback / adjustment: Each student gets one 'focus for planning' tied to a weaker strand.

Differentiation

Representation: Provide the four-strand chain as a labelled visual with icons.

Action & expression: Allow students to complete steps as a spoken chain recorded by a partner.

Engagement: Frame as 'putting on all four hats' of an AI builder.

People of Determination: Give a four-step chain organiser with sentence frames; allow extra time and a scribe.

EAL: Bilingual glossary (system type, prompt, triage, risk check) and linking stems (so, because, which means).

Stretch / next tier: Ask students to explain how changing the prompt would have changed both the output and the risk.

Answer key / model responses

Task / item	Model response	Marking note
Name the system type	A learning system - it finds patterns in language data.	Should contrast with rule-

for a text summariser.		based.
Explain how the prompt shaped the output.	A specific prompt gave a focused, on-task summary; a vague one would drift.	Must link prompt wording to output quality.
Triage the curated output.	Correct, incorrect, or unclear with a checkable reason.	Reason ties to task or source.
Name one responsible-use choice made.	Kept all personal data out of the prompt.	Choice should be made during the task, not after.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Connects the strands	Links system and prompt to the output clearly.	Links two strands only.	Strands treated separately.
Triages the output	Gives a verdict with a checkable reason.	Gives a verdict without a clear reason.	No triage attempted.
Names a responsible-use choice	States a genuine risk-reduction choice.	States a vague choice.	No choice named.

Resources & links

- Four-strand integration challenge pack - Teacher-prepared task, prompt options, and curated output. (teacher-prepared)
- Project success criteria sheet - Criteria linking all four strands for the signature solution. (teacher-prepared)

Week 6 - Lab | Integrated Build-and-Check (Lab)

Strand: Integration | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Lab)

Objectives & success criteria

- Run a small integrated build-and-check task end-to-end and record the checks made.
- Identify which strand most needs work before the signature solution.
- Evaluate the final output for accuracy and risk before deciding to use it.

Student-facing success criteria:

- I can run a build-and-check task through all four strands end-to-end.
- I can record the accuracy and risk checks I made.
- I can name which strand I most need to improve.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Core lesson you connected the four strands into one chain. In the Lab you run a full build-and-check task end-to-end, exactly like the signature solution you will meet next.

Activate: Show the build-and-check board: four stations matching the four strands. Ask what evidence each station needs.

Teach: Explain the build-and-check record and how the final output earns an accuracy and risk verdict; model one station.

Apply: Groups complete the first station (system type) as a warm-up.

Check: Each group states the evidence they will carry to the next station.

Critical & analytical thinking focus

Synthesise - connect all four strands (Analyse/Create | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A build-and-check task mirrors how a real AI solution is produced and then defended. You build by choosing a system suited to the task and designing a prompt that steers it toward a useful output. You check by triaging that output for accuracy and by running a risk assessment before deciding whether to use it. The two halves are equally important: building without checking produces confident mistakes, while checking without building produces nothing. Running the task end-to-end, and recording the evidence at each stage, lets you see which strand is your weakest link - the one most likely to fail under the pressure of the signature solution. Because every step is teacher-mediated and uses curated or anonymised material, you rehearse the full workflow without ever exposing personal data. This is the rehearsal that turns four learned strands into one reliable habit you can trust on a task that matters.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Set up the build-and-check 8 min	Present four stations matching the strands and the build-and-check record. Model station one and the evidence it produces.	Form groups, read the record, note the evidence needed at each station.	• What does build mean here, and what does check mean? • What evidence proves each step?
Build - system and prompt 12 min	Guide groups through stations 1-2: name the system type, then design a clear prompt for the curated task.	Record the system type and their prompt, with a reason for each choice.	• Why is this system suited to the task? • What makes your prompt specific?
Check - accuracy and risk 12 min	Guide groups through stations 3-4: triage the curated output and run a risk assessment, giving the output an accuracy and a risk verdict.	Record a triage verdict with reason and a risk verdict with a mitigation.	• Is the output correct, incorrect, or unclear? • What is the biggest risk, and how would you reduce it?
Decide and diagnose 8 min	Ask each group to decide use / do not use / use-with-changes, then diagnose which strand was their weakest link.	Make the use decision and name the strand to improve.	• Would you use this output as-is? • Which strand let you down most?

Reflect and feedforward 5 min	Collect records; give each group one target strand for the signature solution.	Write their focus strand and one action to strengthen it.	• What will you practise before the big task?

Anticipated misconceptions

Misconception	How to address it
Building the solution is the whole task.	Half the marks are in the check. Model that an unchecked output cannot be used; add the accuracy and risk verdicts.
A use decision is just yes or no.	Show the third option: use-with-changes. Model turning a risky output into a usable one with a fix.
My weakest strand does not matter if the output looks fine.	The weakest link fails under pressure. Diagnose it now so the signature solution does not expose it later.

Hands-on activity guide

Overview: A full end-to-end build-and-check task run across four stations, producing an accuracy verdict, a risk verdict, a use decision, and a diagnosis of the group's weakest strand - a direct rehearsal for the signature solution.

Grouping / timing: Groups of 3-4 moving through 4 stations; individual reflection at the end.

Tool: Teacher-mediated board demo with curated output; anonymised, account-free materials. - Curated task and output supplied by the teacher; students never enter personal or identifying data.

Facilitation steps:

1. Station 1: name the system type suited to the task and why.
2. Station 2: design a clear, specific prompt.
3. Station 3: triage the curated output for accuracy with a reason.
4. Station 4: run a risk assessment and choose a mitigation.
5. Give the output an accuracy verdict and a risk verdict.
6. Decide: use, do not use, or use-with-changes.
7. Diagnose the group's weakest strand before the signature solution.

Unplugged fallback: Unplugged: run all four stations with printed task, prompt options, curated output card, and a risk sheet - no device needed.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Integrates all Term-1 Emerging outcomes end-to-end as a rehearsal for the signature solution

• Check: Completed build-and-check record with accuracy verdict, risk verdict, use decision, and weakest-strand diagnosis

• Success criteria: Runs all four strands end-to-end, records both checks, makes a justified use decision, and names a weakest strand.

• Evidence: Build-and-check record.

• Feedback / adjustment: For build-heavy groups, weight attention to the checks; give each group one target strand for the signature solution.

Differentiation

Representation: Provide a station map with the four strands as labelled icons and a worked example.

Action & expression: Allow the record to be completed with tokens or short phrases rather than full sentences.

Engagement: Frame as a dress rehearsal for the signature solution.

People of Determination: Assign station roles and give a numbered evidence checklist; allow extra time and a scribe.

EAL: Bilingual glossary (build, check, accuracy, risk, mitigation, verdict) and decision stems.

Stretch / next tier: Require a written justification of the use-with-changes option and a redesigned prompt that lowers the risk verdict.

Answer key / model responses

Task / item	Model response	Marking note
Station 1: choose the system type for a summarising task.	A learning system that finds language patterns; suited because summarising needs pattern recognition, not fixed rules.	Choice must be justified by the task.
Station 3: give an accuracy verdict.	Correct / incorrect / unclear with a reason checked against the source.	Reason must be checkable.
Station 4: give a risk verdict and mitigation.	Main risk = using an unverified detail; mitigation = verify before use.	Mitigation must match the risk.
Diagnose the weakest strand.	Any strand, justified by where evidence was thinnest.	Diagnosis should be honest and specific.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Runs the task end-to-end	Completes all four strands with recorded evidence.	Completes most strands with some evidence.	Task not run end-to-end.
Records both checks	Gives a justified accuracy verdict and risk verdict.	Gives one check clearly, the other weakly.	Checks missing or unjustified.
Decides and diagnoses	Makes a justified use decision and names a weakest strand.	Decides or diagnoses, not both.	Neither a decision nor a diagnosis.

Resources & links

- Build-and-check station pack - Teacher-prepared four-station task with a curated output and risk sheet. (teacher-prepared)
- Signature solution readiness checklist - Checklist mapping each strand to the evidence needed. (teacher-prepared)

Week 7 - Core | Signature Solution - Planning

Strand: Integration (Signature Solution - Planning) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Core)

Objectives & success criteria

- Plan a structured AI solution: state the goal, the prompt structure, and the check steps.
- Identify the inputs/data and the output-evaluation method.
- Describe one risk and how the plan reduces it.

Student-facing success criteria:

- I can state the goal of my solution in one clear sentence.
- I can lay out a prompt structure with a role, a task, and one rule.
- I can name the inputs my solution needs and how I will check the output.
- I can describe one risk and one way my plan reduces it.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You have used prompts before and know that AI systems can be rule-based or learning-based. In Grade 5 you improved a single prompt. Now you plan a whole solution: not just one prompt, but a structure with inputs, checks and a fallback.

Activate: Show two board demos side by side: a vague request ('Help with the trip') and a structured one ('You are a bus timetable helper. List the three earliest buses from Al Ain to Abu Dhabi after 7am. If a time is missing, say so.'). Ask which is easier to check and why.

Teach: Explain that a signature solution is a small, reusable design: a goal, a prompt structure (role + task + one rule), the inputs it needs, an evaluation step, and a risk check. Model naming each part on the board.

Apply: In pairs, students label the parts of the structured demo prompt (role, task, rule) and suggest one check step for its output.

Check: Cold-call two pairs: 'What is the goal in one sentence?' and 'How would you check the output is correct?'

Critical & analytical thinking focus

Design & plan - architect a prompted solution (Create (plan) | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A structured, prompt-driven solution is more than a single question typed into a tool. It is a small system with parts that fit together. First comes the goal: one plain sentence that says exactly what the solution should do and for whom. Next comes the prompt structure, which usually names a role for the AI, a clear task, and at least one rule or limit that keeps the output focused. The solution also needs defined inputs, which is the information the prompt works on, and a matching output, which is the result you can inspect. Because AI systems, whether rule-based or learning-based, can produce wrong or unclear answers, a good plan builds in an evaluation step that says how you will judge the output for accuracy. A strong plan also names one realistic risk, such as a confident but incorrect answer, and shows one way the design lowers that risk, like adding a verification rule or a human check. Planning this way is systems thinking: you decide how the pieces connect before you build, so that problems are cheaper to catch and fix.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Hook and goal-setting 7 min	Present the vague vs structured board demo and define 'signature solution' as a reusable design with parts. Reveal the Emerging project criteria.	Compare the two demos and write, in one sentence, what a good solution goal should include.	• What makes a request easy or hard to check? • What does a goal sentence need to be useful?
Model the plan parts 10 min	Think aloud while building a sample plan for a simple task (e.g. sorting lost-property notices), naming goal, prompt structure, inputs, evaluation, and one risk.	Annotate the modelled plan on their sheet, marking each of the five parts.	• Which part is the role, task, and rule? • What input does this prompt need to work?

Guided planning 12 min	Hand each pair a task card. Circulate and prompt for the evaluation and risk steps, which are the most-often missed.	Draft the goal, prompt structure, and inputs for their own task card.	• Is your goal one clear sentence? • What exactly will you feed in as input?
Add evaluation and risk 11 min	Model one evaluation method (compare output to a known correct answer) and one risk mitigation (add a 'say if unsure' rule). Support POD and EAL pairs with sentence frames.	Write how they will check the output and name one risk plus one way the plan reduces it.	• How will you know the output is accurate? • What could go wrong, and how does your plan help?
Plan review and approve-to-build gate 5 min	Run a quick plan review against the rubric; approve plans that are complete or return them with one targeted prompt.	Self-check the plan against the five-part list and swap for a peer glance before submitting.	• Which of the five parts is strongest? • Which part still needs one more sentence?

Anticipated misconceptions

Misconception	How to address it
A good prompt is just a longer question.	Length is not structure. Show that a short prompt with a role, a task, and one rule is easier to check than a long, rambling one. Structure beats size.
If the AI sounds confident, the answer must be correct.	Explain that learning systems can be confidently wrong. That is exactly why the plan needs an evaluation step and a verification rule.
Planning is a waste of time - just start typing prompts.	Show that a plan makes mistakes cheaper to catch. Naming inputs, checks, and risks before building saves rework and keeps data safe.

Hands-on activity guide

Overview: This Core lesson is planning-focused rather than a full build; the hands-on work is producing an approved plan sheet for the signature solution.

Grouping / timing: Pairs, with a teacher-led plan-review gate at the end.

Tool: No live AI account for students. Teacher board demo plus curated example outputs only. - Teacher-mediated on the board using a school-approved child-safe tool; students do not log in.

Facilitation steps:

1. Read your task card and write the goal in one sentence.
2. Build the prompt structure: role, task, and one rule.
3. List the inputs the prompt needs (no personal or identifying data).
4. Write your output-evaluation method (how you will check accuracy).
5. Name one risk and one way your plan reduces it, then submit for the approve-to-build gate.

Unplugged fallback: If no tool is available, work entirely on planning cards using curated sample outputs pinned to the board; the plan is complete without any live AI.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Checks planning of the signature solution against the Emerging project criteria
• Check: Plan review against the project rubric (Emerging); approve-to-build gate
• Success criteria: Plan states goal, prompt structure, inputs, an evaluation method, and one risk mitigation.
• Evidence: Approved plan sheet.
• Feedback / adjustment: Plans missing an evaluation or risk step are returned with one targeted prompt.

Differentiation

Representation: Offer the plan as a labelled diagram and as a five-line checklist so learners can choose a visual or a text route.

Action & expression: Students may record the plan by writing, by placing pre-made card labels, or by dictating to a partner scribe.

Engagement: Task cards use familiar UAE contexts (school lost-property, majlis event list, dates-market stall) so the goal feels real.

People of Determination: Provide a partly completed plan frame and enlarged, high-contrast cards; allow extra think time and a quiet planning corner.

EAL: Give sentence frames: 'The goal is to ___.', 'The AI acts as a ___.', 'One rule is ___.', 'I will check the output by ___.', 'One risk is ___; my plan reduces it by ___.'

Stretch / next tier: Add a second rule and a second evaluation method, and explain which check catches which kind of error.

Answer key / model responses

Task / item	Model response	Marking note
Write the goal in one sentence for a lost-property sorter.	The goal is to sort short lost-property notices into 'clothing', 'devices', or 'other' so staff can return items faster.	A strong goal names the task and who it helps.
Build a prompt structure for that goal.	You are a lost-property sorting helper. Read each notice and label it clothing, devices, or other. If a notice fits none, label it 'other' and say why.	Role + task + one rule; no names or personal data included.
State the evaluation method.	I will compare ten labelled notices to a teacher answer key and count how many labels match.	Accuracy is checked against a known-correct reference.
Name one risk and a mitigation.	Risk: the AI labels an unclear notice confidently but wrongly. Mitigation: the rule forces it to say 'other' and give a reason, and a person checks any 'other'.	The mitigation connects directly to the named risk.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Completeness of plan (five parts)	Plan clearly states goal, prompt structure, inputs, an evaluation method, and one risk mitigation.	Plan includes most parts but one (often evaluation or risk) is thin or missing.	Plan is a loose prompt with two or fewer parts named.
Quality of evaluation and risk thinking	Evaluation method is checkable and the mitigation clearly reduces the named risk.	Evaluation or mitigation is present but vague or only loosely linked to a risk.	No workable evaluation method or risk is identified.
Responsible-use and data safety	Plan uses no personal or identifying data and includes a human check.	Plan mostly avoids personal data but the human check is unclear.	Plan would put personal or identifying data into the prompt.

Resources & links

- Signature Solution Plan Sheet (Emerging) - One-page five-part planning frame with sentence frames and a self-check strip. (None)
- Curated Output Cards - Set A - Teacher-selected safe sample outputs used for evaluation practice, no student data. (None)

Week 7 - Lab | Prototype and Pressure-Test the Plan (Lab)

Strand: Integration (Signature Solution - Planning) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Lab)

Objectives & success criteria

- Prototype the prompt structure and pressure-test the plan against a tricky input.
- Add a fallback branch for when the output fails a check.
- Record what changed and why after the pressure test.

Student-facing success criteria:

- I can run my prompt structure on a normal input and read the output.
- I can pressure-test it with a tricky input and spot where it fails.
- I can add a fallback branch that says what to do when a check fails.
- I can record the change I made and the reason for it.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You have an approved five-part plan from the Core lesson. Today you turn the plan into a working prototype on the board and stress it with hard inputs, then add a fallback so failures are handled, not hidden.

Activate: Recap the approved plan structure. Show a curated output where a normal input works, then ask: 'What if the input is messy or missing?'

Teach: Define prototype (a first working version), pressure test (trying a tricky input on purpose), and fallback branch (a planned response when a check fails).

Apply: In pairs, students predict how their prompt would behave on one tricky input before it is tested on the board.

Check: Ask two pairs to state their tricky input and what a good fallback would say.

Critical & analytical thinking focus

Prototype & test - stress a design and harden it (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

A prototype is the first working version of a solution, built quickly so you can see how it really behaves rather than how you hoped it would. Pressure-testing means deliberately feeding a design its hardest inputs: a missing value, an unclear notice, or two items in one line. These edge cases reveal weak spots that normal inputs hide. When an output fails a check, a well-designed solution does not simply stop or give a wrong answer confidently; it follows a fallback branch, which is a planned response such as 'if the item is unclear, label it other and ask a person'. Designing fallbacks is a form of defensive design used in real engineering, from lifts to payment apps. In a teacher-mediated classroom, the AI runs only on the board using a school-approved child-safe tool, and no personal or identifying data is ever entered, so pressure-testing uses safe, curated inputs. Each test is logged so the class can see the pattern of failures and decide which single change will help the most. This is systems thinking in action: you improve the design, not just one answer.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Recap and set the build target 6 min	Recap the approved plan and set today's target: prototype, pressure-test, add one fallback branch.	Re-read their plan and underline the one rule most likely to fail under a tricky input.	• What is your prompt supposed to do on a normal input? • Where might it break?
Prototype on a normal input 10 min	Run two or three pairs' prompt structures on the board with a normal curated input; capture outputs on the build/test log.	Read the output and mark whether it met the goal, recording the result on their log.	• Did the output meet the goal? • Is anything unclear or extra?
Pressure-test with tricky	Feed each prototype a curated tricky input (missing, messy, or double item) and lead	Record what failed and classify the error type	• What kind of error is this? • Did the AI sound confident even

inputs 14 min	evaluation of the failure.	(wrong, unclear, or missing).	when wrong?
Add a fallback branch 10 min	Model writing an 'if a check fails, then…' branch and adding it to the prompt structure; support POD/EAL with a branch frame.	Write a fallback branch for their own solution and re-run it on the tricky input if time allows.	• What should happen when the check fails? • Does your fallback keep a person in the loop?
Log the change and reason 5 min	Collect build/test logs and confirm each shows one change with a reason.	Complete the 'change made / reason' line and self-rate readiness to build.	• What one change did you make? • Why will it help next week's build?

Anticipated misconceptions

Misconception	How to address it
If it works on a normal input, the solution is finished.	Normal inputs hide weak spots. Show a prototype that passes a normal case but fails a missing value; the tricky input is the real test.
A fallback means the solution failed.	A fallback is good design, not failure. It turns a silent wrong answer into a safe, handled outcome with a person in the loop.
We can pressure-test using real names or student data to be realistic.	Never. Realism does not need personal data. Use curated, made-up inputs; entering identifying data is against the rules and unsafe.

Hands-on activity guide

Overview: A substantial prototyping session: students turn their approved plan into a working prompt structure, stress it with curated tricky inputs on the board, and design a fallback branch, all logged.

Grouping / timing: Pairs at planning stations feeding the teacher-run board demo in a rotation.

Tool: School-approved child-safe AI tool run by the teacher on the board; simulated-AI cards as backup. - Teacher-mediated only; no student logins. All inputs are curated and contain no personal or identifying data.

Facilitation steps:

1. Copy your approved prompt structure onto the prototype slot of your build/test log.
2. Run it on a normal curated input (teacher board) and record whether it met the goal.
3. Pressure-test with a tricky curated input; record what failed and the error type.
4. Write a fallback branch: 'If the check fails, then ___ and ask a person.'
5. Re-run on the tricky input if time allows and log the change you made and the reason.

Unplugged fallback: Unplugged version: lay the plan out on cards and use curated output cards as the 'AI response'; a partner reads the card so pairs can pressure-test and design a fallback with no live tool.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Checks that the plan survives a tricky input and gains a fallback branch

• Check: Build/test log review: one normal run, one pressure test, one fallback branch, one logged change

• Success criteria: Prototype is run, a tricky input exposes a weak spot, and a fallback branch plus a logged reason are added.

• Evidence: Completed build/test log with fallback branch.

• Feedback / adjustment: Logs without a fallback or without a classified error type are returned with one targeted prompt.

Differentiation

Representation: Show the fallback as a branching flow arrow and as an 'if-then' sentence so both visual and text learners can follow.

Action & expression: Students can prototype by writing, by arranging card labels, or by narrating their run to a scribe.

Engagement: Tricky inputs use playful but safe UAE contexts (a smudged majlis guest note, a two-items-in-one lost-property line).

People of Determination: Provide a pre-drawn branch template and enlarged log; allow a quieter station and extra processing time.

EAL: Give the frame 'If ___ then ___ and ask a person.' plus an error-type word bank: wrong, unclear, missing.

Stretch / next tier: Design two fallback branches for two different failure types and explain which is more likely and why.

Answer key / model responses

Task / item	Model response	Marking note
Prototype run on a normal input.	Input 'blue school jumper found in hall' -> output 'clothing'. Meets the goal.	Normal case passes; this is the baseline, not proof of a finished solution.
Pressure test with a tricky input.	Input 'found: charger and a scarf' -> output 'devices' only. Error type: missing (it dropped the scarf).	The tricky input exposes a real weak spot to fix.
Write a fallback branch.	If a notice names two items, then label each item separately, and if still unclear label 'other' and ask a person.	The branch handles the exposed failure and keeps a human in the loop.
Log the change and reason.	Change: added a two-item rule and a fallback. Reason: the pressure test showed the prompt dropped the second item.	The reason connects directly to the observed failure.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Prototyping and testing	Runs the prototype on a normal input and pressure-tests it with a tricky input, recording both.	Runs one case but the pressure test or the recording is incomplete.	No working prototype run is recorded.
Fallback design	Adds a clear if-then fallback branch that handles the exposed failure and keeps a person in the loop.	Adds a fallback but it is vague or does not match the failure.	No fallback branch is added.
Evidence and reasoning	Logs the change with a reason that links to the observed failure and classifies the error type.	Logs a change but the reason or error type is missing.	No change or reasoning is logged.

Resources & links

- Build/Test Log (Lab) - Two-column log for prototype runs, pressure tests, fallback branch and logged change. (None)
- Curated Tricky-Input Cards - Teacher-selected safe edge-case inputs (missing, messy, double item), no personal data. (None)

Week 8 - Core | Signature Solution - Building

Strand: Integration (Signature Solution - Building) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Core)

Objectives & success criteria

- Build the solution by following the plan and prompt structure.
- Evaluate outputs for accuracy and flag any that are incorrect or unclear.
- Make at least one improvement based on an output check.

Student-facing success criteria:

- I can build my solution by following my approved plan.
- I can evaluate outputs and flag ones that are wrong or unclear.
- I can make one improvement and explain what changed and why.
- I can keep personal data out of every prompt.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You have an approved plan and a pressure-tested prototype with a fallback branch. Today you build the real solution by following that plan, evaluate the outputs, and make one improvement based on what you find.

Activate: Show a curated set of five outputs from a sample solution; two are subtly wrong. Ask students to flag the wrong ones.

Teach: Explain build discipline: follow the plan, run inputs, evaluate each output against the check, flag errors, then change one thing. Define 'evaluate -> change -> result'.

Apply: Pairs practise flagging by marking the two wrong sample outputs and saying why.

Check: Ask two pairs which output they flagged and how they knew it was wrong.

Critical & analytical thinking focus

Create & evaluate - build, evaluate, iterate (Create/Evaluate | DOK 3 | SOLO: Relational)

The technical concept (teacher background)

Building a signature solution means following your plan with discipline rather than improvising new prompts each time. You feed defined inputs into your prompt structure, capture the outputs, and evaluate each one against the check you designed: does it match the known-correct answer, and is it clear? Because learning-based AI can be confidently wrong, evaluation is not optional; it is the step that separates a working solution from a risky one. When you find an output that is incorrect or unclear, you flag it, which means marking it for attention rather than quietly accepting it. Then you make one improvement, usually a small change to a rule or the prompt structure, and you record the loop as 'evaluate -> change -> result'. This iteration loop is how real systems mature: one measured change, one observed result. Throughout the build you keep personal and identifying data out of every prompt and keep a person in the loop for anything flagged. A single, well-explained improvement backed by evidence is worth more than many untested changes.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Build launch and gate check 6 min	Confirm each pair has an approved plan and fallback; set the build target and the evaluate-change-result expectation.	Set up their build/test log and restate their goal and check in one line.	• What is your check for a correct output? • What is your fallback if it fails?
Build by following the plan 12 min	Run pairs' prompt structures on the board with defined inputs; keep the build faithful to the plan.	Record each output on the log without changing the prompt yet.	• Are you following the plan or improvising? • Is the input free of personal data?
Evaluate and flag 12 min	Lead evaluation of outputs against the check; model flagging an incorrect or unclear one.	Mark each output correct, unclear, or wrong and flag any that fail the check.	• Does this output match the known-correct answer? • Is it clear enough to act on?

Make one improvement 10 min	Model changing one rule in response to a flagged output and re-running it; support POD/EAL with an improvement frame.	Make one improvement, re-run, and log evaluate -> change -> result.	• What one change addresses the flag? • Did the result actually improve?
Improvement share and check-in 5 min	Run a quick roving check against the rubric; pair stuck students with a buddy and a stronger prompt structure.	Share their one logged improvement with another pair.	• What did your change fix? • What would you improve next?

Anticipated misconceptions

Misconception	How to address it
Building means writing new prompts until something looks good.	That is guessing, not building. Following the approved plan and changing one thing at a time lets you see what actually caused an improvement.
If most outputs look right, I do not need to check each one.	Subtle errors hide among correct-looking outputs. Evaluate every output against the check; that is how you catch the confident wrong answer.
More changes at once means faster improvement.	Many changes at once hide which one helped. One change plus one observed result gives evidence you can trust.

Hands-on activity guide

Overview: Core build session: students build the solution from their approved plan, evaluate outputs, flag errors, and log one evidence-based improvement.

Grouping / timing: Pairs feeding the teacher-run board demo in rotation, with a buddy system for stuck pairs.

Tool: School-approved child-safe AI tool run by the teacher on the board; simulated-AI cards as backup. - Teacher-mediated only; students never log in and never enter personal or identifying data.

Facilitation steps:

1. Restate your goal and check on the build/test log.
2. Run your prompt structure on defined, curated inputs (teacher board) and record each output.
3. Evaluate each output against your check and flag any that are wrong or unclear.
4. Make one improvement to a rule or the structure in response to a flag.
5. Re-run and log the loop as evaluate -> change -> result.

Unplugged fallback: Unplugged: use curated output cards as the AI responses; pairs evaluate and flag on the cards, then rewrite one rule and predict the new result.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Monitors build against the plan; each student logs one evaluation-based improvement

• Check: Build check-in against plan + one logged improvement ('evaluate -> change -> result')

• Success criteria: Solution follows the plan, outputs are evaluated, and one improvement is made and explained.

• Evidence: Build progress + improvement log.

• Feedback / adjustment: Roving feedback against the rubric; stuck students get a stronger prompt structure and a peer buddy.

Differentiation

Representation: Present the iteration loop as a cycle diagram and as a three-word phrase (evaluate, change, result).

Action & expression: Students may record outputs by writing, ticking a flag column, or narrating to a scribe.

Engagement: Inputs stay in familiar UAE settings (school events, souq stall, library returns) so evaluation feels purposeful.

People of Determination: Provide a pre-formatted log with a flag column and enlarged text; allow extra time and a calm build station.

EAL: Give frames: 'This output is correct / unclear / wrong because ___.' and 'I changed ___ so that ___; the result was ___.'

Stretch / next tier: Make a second improvement and compare which change gave the bigger accuracy gain, with evidence from the log.

Answer key / model responses

Task / item	Model response	Marking note
Evaluate an output against the check.	Output 'clothing' for 'blue jumper' matches the answer key, so it is correct.	Evaluation compares to a known-correct reference, not to a hunch.
Flag an incorrect output.	Output 'other' for 'tablet charger' is wrong; it should be 'devices'. Flag it.	Flagging marks it for a change and a human check.
Make one improvement.	Added a rule: 'chargers, cables and headphones count as devices'. Re-ran and the charger was labelled correctly.	One change, then one observed result.
Log evaluate -> change -> result.	Evaluate: charger mislabelled. Change: added a devices examples rule. Result: charger now 'devices', accuracy up.	The loop shows cause and effect clearly.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Faithful build	Builds the solution by following the approved plan and prompt structure with defined inputs.	Builds but drifts from the plan or improvises prompts in places.	No plan-based build is produced.
Evaluation and flagging	Evaluates outputs against the check and flags incorrect or unclear ones accurately.	Evaluates some outputs but misses or mis-flags errors.	Outputs are accepted without evaluation.
Evidence-based improvement	Makes and explains one improvement with a clear evaluate -> change -> result log.	Makes a change but the reason or result is unclear.	No improvement or explanation is logged.

Resources & links

- Build/Improvement Log (Core) - Log with input, output, flag, and an evaluate-change-result strip. (None)
- Curated Output Set - Two Errors - Teacher-selected outputs with two seeded errors for flagging practice, no student data. (None)

Week 8 - Lab | Extended Build and Evaluation (Lab)

Strand: Integration (Signature Solution - Building) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Lab)

Objectives & success criteria

- Complete a working solution and evaluate its outputs across several test inputs.
- Handle an incorrect output gracefully and document the fix.
- Run a basic risk check and confirm no personal data is used.

Student-facing success criteria:

- I can complete a working solution that follows my plan.
- I can evaluate its outputs across several test inputs and measure accuracy.
- I can handle an incorrect output gracefully using my fallback.
- I can run a basic risk check and document the fix I made.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: You have built the first version and logged one improvement. Today you complete the solution, evaluate it across several test inputs, handle an incorrect output with your fallback, and run a basic risk check before you present it.

Activate: Show a nearly finished sample solution and a test set of six curated inputs; ask how we would prove it is accurate.

Teach: Define a test set, an accuracy count (correct out of total), graceful handling (using the fallback), and a basic risk check (accuracy, clarity, no personal data).

Apply: Pairs decide what counts as a correct output before testing, so evaluation is fair.

Check: Ask two pairs to state their accuracy target and their risk-check questions.

Critical & analytical thinking focus

Complete & evaluate - finish, test across cases, harden (Create/Evaluate | DOK 4 | SOLO: Extended abstract)

The technical concept (teacher background)

Completing a solution means more than getting one good answer; it means proving the solution behaves well across a set of test inputs. A test set is a small, curated collection of inputs, including a few edge cases, that you run through the finished solution to measure accuracy as a simple count of correct outputs out of the total. When an output is incorrect, a completed solution handles it gracefully by following its fallback branch rather than presenting a confident wrong answer, and you document the fix so the change is traceable. A basic risk check asks three questions: is the solution accurate enough for its purpose, are the outputs clear enough to act on, and does any prompt contain personal or identifying data. In a teacher-mediated setting the AI runs only on the board with curated inputs, so the data-safety answer must always be that no personal data is used. This is DOK 4 work because you are integrating build, evaluation across cases, fallback handling, and a risk judgement into one coherent product, and reasoning about how the parts affect each other. A solution you can measure and defend is far more trustworthy than one that merely looks finished.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Set the completion target and test set 6 min	Confirm each pair's fallback works; introduce the curated test set of six inputs and define a correct output.	Write their accuracy target and what counts as a correct output.	• What does a correct output look like? • How many correct out of six is good enough?
Complete the build 10 min	Run each pair's solution to a finished state on the board, ensuring the fallback branch is wired in.	Confirm their solution follows the plan and the fallback is present, recording readiness.	• Is the fallback branch actually in the prompt? • Does the build match the plan?
Evaluate across	Run all six curated inputs; lead a count of	Record each result, mark	• Where did accuracy drop?

the test set 13 min	correct outputs and identification of any failure.	correct or not, and compute accuracy (correct out of six).	• Did an edge case trigger the fallback correctly?
Handle a failure and run the risk check 11 min	Model graceful handling of one incorrect output via the fallback, then lead the three-question risk check; support POD/EAL with frames.	Document the fix and complete the accuracy / clarity / no-personal-data risk check.	• Did the fallback handle the failure safely? • Is any prompt free of personal data?
Showcase and reflect 5 min	Invite two pairs to show their accuracy count and risk check; confirm evidence against the rubric.	Share their accuracy result and one documented fix.	• How accurate is your solution and how do you know? • What is the biggest remaining risk?

Anticipated misconceptions

Misconception	How to address it
One good output means the solution is accurate.	Accuracy is about the whole test set, not one lucky case. Count correct outputs across several inputs to make a fair claim.
Handling a failure means deleting the wrong output and moving on.	Graceful handling means the fallback catches it and a person is involved, and the fix is documented so it is traceable.
A risk check is only about accuracy.	A basic risk check also asks about clarity and data safety. A prompt with personal data is a serious risk even if the answer is accurate.

Hands-on activity guide

Overview: Extended build and evaluation: students complete the solution, run it across a curated six-input test set, measure accuracy, handle a failure with the fallback, document the fix, and complete a basic risk check.

Grouping / timing: Pairs feeding the teacher-run board demo in rotation, with a showcase at the end.

Tool: School-approved child-safe AI tool run by the teacher on the board; simulated-AI cards as backup. - Teacher-mediated only; no student logins; all six test inputs are curated and contain no personal or identifying data.

Facilitation steps:

1. Confirm your solution follows the plan and the fallback branch is wired in.
2. Define what counts as a correct output for your goal.
3. Run all six curated test inputs on the board and record each result.
4. Compute accuracy (correct out of six) and handle any failure with your fallback.
5. Document the fix and complete the three-question risk check (accuracy, clarity, no personal data).

Unplugged fallback: Unplugged: use six curated output cards as the AI responses; pairs evaluate accuracy, act out the fallback for a failing card, and complete the same risk check without any live tool.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Confirms a completed solution is evaluated across several inputs, handles a failure, and passes a basic risk check

• Check: Test-set evaluation log (accuracy count) + documented fix + completed risk check

• Success criteria: Solution is completed, evaluated across several inputs with an accuracy count, a failure is handled gracefully, and the risk check confirms no personal data.

• Evidence: Completed test-set log, documented fix, and risk-check sheet.

• Feedback / adjustment: Logs without an accuracy count or a completed risk check are returned with one targeted prompt.

Differentiation

Representation: Show accuracy as a simple count and as a filled bar; show the risk check as a three-tick card.

Action & expression: Students may tally accuracy in writing, with counters, or by narrating to a scribe.

Engagement: The test set uses varied UAE contexts (Eid event notices, souq stock lines, sports-day sign-ups) to keep evaluation lively and fair.

People of Determination: Provide a pre-formatted test-set log and risk-check card with enlarged text; allow extra time and a calm station.

EAL: Give frames: 'My solution got ___ out of 6 correct.', 'The fallback handled ___ by ___.', 'Risk check: accurate? ___ clear? ___ personal data used? no.'

Stretch / next tier: Add a seventh, harder edge case, predict the result, then test and explain whether your solution generalised.

Answer key / model responses

Task / item	Model response	Marking note
Compute accuracy across the test set.	Five of six outputs matched the answer key, so accuracy is 5/6.	Accuracy is a count across the whole set, not one case.
Handle an incorrect output gracefully.	The two-item notice failed, so the fallback split the items and flagged the unclear one for a person. The wrong output was not presented as final.	Graceful handling uses the fallback and keeps a person in the loop.
Document the fix.	Fix: strengthened the two-item rule and confirmed the fallback triggers on unclear notices; re-ran and accuracy rose to 6/6.	The fix is traceable and its effect is recorded.
Run the basic risk check.	Accurate? mostly (6/6 after fix). Clear? yes. Personal data used? no - all inputs curated. Biggest remaining risk: rare unseen edge cases.	The risk check covers accuracy, clarity, and data safety.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Completed, plan-faithful solution	Solution is completed, follows the plan, and has the fallback branch wired in.	Solution is mostly complete but the plan or fallback is partly missing.	Solution is not completed to a working state.
Evaluation across inputs	Evaluates outputs across the test set and reports an accuracy count with a failure handled gracefully.	Evaluates some inputs but the accuracy count or failure handling is incomplete.	No multi-input evaluation is recorded.
Risk check and data safety	Completes the three-question risk check and confirms no personal or identifying data is used.	Completes part of the risk check or the data-safety confirmation is unclear.	No risk check is completed, or personal data would be used.

Resources & links

- Test-Set Evaluation Log (Lab) - Six-row log with judgement and accuracy count, plus a documented-fix strip. (None)
- Basic Risk-Check Card - Three-question card: accuracy, clarity, no personal data, with a remaining-risk line. (None)

Week 9 - Core | Signature Solution - Refining & Finishing

Strand: Integration (Signature Solution - Refining & Finishing) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Core)

Objectives & success criteria

- Finish the solution so it meets the agreed success criteria.
- Evaluate it for accuracy, usefulness and risk.
- Explain one change that improved it.

Student-facing success criteria:

- I can finish my solution so it meets the success criteria our class agreed on.
- I can judge my solution for accuracy, usefulness and risk using the rubric.
- I can explain one change I made and say why it made the solution better.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In Weeks 5-8 you designed and built a signature solution that uses rule-based and learning ideas to help a real user at your school in Abu Dhabi. You already know that a rule-based system follows fixed rules a person writes, while a learning system finds patterns in example data. You have tested an early version and collected some feedback.

Activate: Show the class the agreed success-criteria card from Week 6 on the board. Ask pairs to whisper one thing their solution already does well and one thing that is still rough.

Teach: Explain that refining means comparing your work to the criteria on purpose, not just guessing. Model one accuracy check, one usefulness check and one risk check on a curated demo solution.

Apply: Pairs mark the demo solution against three rubric rows (accuracy, usefulness, risk) and agree the single most important improvement.

Check: Cold-call two pairs to name the improvement they chose and the evidence behind it.

Critical & analytical thinking focus

Evaluate & justify - quality and risk against criteria (Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

Refining an AI-supported solution means measuring it against the success criteria you agreed earlier, not against how you feel about it. A strong evaluation looks at three different things at once. Accuracy asks how often the system gives the right or sensible answer, and you check it by testing with examples where you already know the correct outcome. Usefulness asks whether the solution actually helps the real user do their task faster, more easily or more fairly. Risk asks what could go wrong, who could be harmed, and whether any private information is exposed. In a rule-based part, refining often means fixing or adding a rule; in a learning part, it often means giving cleaner or more balanced example data. A single well-chosen improvement, backed by evidence, is worth more than many random changes. Responsible builders always verify a change actually helped by re-testing, because a change that fixes one case can quietly break another.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Connect 7 min	Recap the success-criteria card and the three evaluation lenses (accuracy, usefulness, risk). Share the lesson goal: finish and improve one thing with evidence.	Recall their own criteria and note where their solution stands.	• What did we agree a good solution must do? • Which lens matters most for your user?
Model 10 min	On a curated demo solution, think aloud through one accuracy test, one usefulness check and one basic risk check. Show how to write a short refinement note.	Follow the model and annotate a copy of the demo rubric.	• How do I test accuracy fairly? • What evidence proves this is useful? • What could go wrong here?

Guided evaluation 10 min	Circulate as pairs run the same three checks on their own solution and record judgements on the tiered rubric.	Self-evaluate the solution against the Emerging rubric and mark accuracy, usefulness and risk.	• Where does my evidence come from? • Which row is not yet at Emerging?
Refine 12 min	Support each pair to choose ONE improvement, make it, and re-test to confirm it helped without breaking something else.	Make one targeted change (fix a rule or improve example data) and re-test.	• Did the change fix the problem? • Did anything else get worse?
Consolidate 6 min	Lead 'two stars and a wish' peer feedback and collect exit notes. Reinforce the responsible-use message about never using real personal data.	Give and receive structured peer feedback; write the refinement note.	• What is one star and one wish for this solution? • Is any private data being used? How do we keep it out?

Anticipated misconceptions

Misconception	How to address it
A change always makes the solution better.	Show that a change must be re-tested. A change can fix one case and break another, so evidence after the change is required.
Refining just means making it look nicer.	Refining is measured against the success criteria for accuracy, usefulness and risk, not only appearance.
If it works on my one example, it is accurate.	Accuracy needs several test cases with known correct answers, including tricky ones, not a single lucky example.

Hands-on activity guide

Overview: Pairs run a structured self-evaluation of their signature solution against the tiered rubric, choose one improvement, apply it, and re-test to confirm it helped. This mirrors the Lab, which goes deeper on the risk review.

Grouping / timing: Same working pairs as the build weeks, with a peer-review swap for feedback.

Tool: Board demo plus printed rubric and refinement-note sheet; school-approved child-safe tool used by the teacher only for any live check. - Account-free for students. Teacher-mediated demo on the class board; no student logins; no personal data entered into any tool.

Facilitation steps:

1. Open the agreed success-criteria card and read the three lenses.
2. Run one accuracy test using examples with known correct answers.
3. Run one usefulness check against what the real user needs.
4. Run one basic risk check (harm, fairness, privacy).
5. Choose the single most important improvement and record why.
6. Make the change and re-test to confirm it helped and broke nothing.
7. Swap with another pair for 'two stars and a wish' feedback.

Unplugged fallback: If no tool or board is available, use the simulated-AI cards: pairs judge printed curated outputs against the rubric and plan the improvement on paper, exactly the same thinking without any device.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Self- and peer-evaluation against the tiered rubric (Emerging = expected)

• Check: Structured self/peer review using the rubric; one improvement acted upon

• Success criteria: Meets the Emerging criteria; names an accuracy/usefulness/risk judgement and one improvement.

• Evidence: Completed self/peer rubric + refinement note.

• Feedback / adjustment: Peer feedback structured ('two stars and a wish'), teacher-checked for accuracy.

Differentiation

Representation: Provide the criteria as a visual card, spoken recap and printed checklist so every learner can access it.

Action & expression: Learners may record judgements by writing, drawing, or speaking to a scribe.

Engagement: Anchor the task in each pair's real Abu Dhabi user so the improvement feels meaningful.

People of Determination: Offer sentence stems, chunked steps, extra think-time and a partner who can scribe; allow a quieter review corner.

EAL: Pre-teach accuracy, useful, risk, improve with bilingual glossary and picture cues; accept first-language notes.

Stretch / next tier: Ask advanced pairs to justify why their chosen improvement beats a second candidate improvement, using evidence.

Answer key / model responses

Task / item	Model response	Marking note
Name one accuracy check for your solution.	I tested it on five cases where I already knew the correct answer and counted how many it got right.	Look for known-answer test cases, not opinion.
State one usefulness judgement.	It helps Aisha sort library returns faster, so it is useful for her real task.	Must link to the real user's task.
Describe one risk you checked.	I checked no student names or photos are used, so private data is protected.	Accept harm, fairness or privacy risks.
Explain one change that improved it.	I added a rule for empty inputs, then re-tested; accuracy went from 3/5 to 5/5 and nothing else broke.	Requires before/after evidence.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Evaluates against criteria (accuracy, usefulness, risk)	Judges all three lenses with evidence from testing.	Judges one or two lenses, evidence partly clear.	Gives opinion without testing evidence.
Acts on one improvement	Makes one targeted change and re-tests to confirm it helped.	Makes a change but does not re-test it.	No clear improvement made.
Responsible use (privacy/fairness)	Confirms no personal data used and considers fairness.	Mentions privacy or fairness but not both.	Does not consider privacy or fairness.

Resources & links

- ADEK K-12 AI Fluency Framework - The four-strand outcomes this unit is built on. (https://www.adek.gov.ae)
- Class success-criteria and rubric card - The agreed Emerging criteria used for self and peer review. (school-shared)

Week 9 - Lab | Final Iteration and Risk Review (Lab)

Strand: Integration (Signature Solution - Refining & Finishing) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Lab)

Objectives & success criteria

- Run a final evaluation across accuracy, usefulness and risk, and act on one improvement.
- Complete a risk review and confirm the solution is safe and fair to use.
- Document the final iteration with before-and-after evidence.

Student-facing success criteria:

- I can run a final evaluation across accuracy, usefulness and risk and act on one improvement.
- I can complete a risk review and confirm my solution is safe and fair.
- I can record my final iteration with clear before-and-after evidence.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Week 9 Core lesson you judged your signature solution with three lenses and made one improvement. In this Lab you carry the same solution and the same rubric into a full final iteration, adding a deeper risk review before you freeze the solution for showcase.

Activate: Pairs re-open their refinement log from Core and read out their last recorded judgement.

Teach: Introduce the risk matrix (how likely x how serious) and model placing one risk on it for a curated demo solution.

Apply: Pairs place one real risk from their own solution onto a printed risk matrix.

Check: Each pair states whether their placed risk needs action now or just watching.

Critical & analytical thinking focus

Evaluate & justify - quality and risk against criteria (Evaluate | DOK 3 | SOLO: Relational to Extended abstract)

The technical concept (teacher background)

A risk review is a structured check of what could go wrong with a solution before real people rely on it. Two questions shape every risk: how likely is it to happen, and how serious would it be if it did. Plotting a risk on a matrix of likelihood against seriousness helps you decide what needs fixing now and what only needs watching. For AI-supported solutions, common risks include a wrong answer that misleads a user, unfair treatment of one group because the example data was unbalanced, and exposure of private information. The most important privacy rule for students is simple: never put personal or identifying data into a prompt or a training set. A final iteration means you make your last planned improvement, re-test with known-answer cases, and confirm the solution is safe and fair before you freeze it. Freezing means you stop changing it so the version you showcase is the version you evaluated. Good documentation of before-and-after evidence lets others trust your judgement.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Connect 6 min	Recap the three lenses and introduce today's focus: a full final iteration plus a deeper risk review.	Re-read their refinement log and set one goal for the Lab.	• Where did we stop last lesson? • What is left to finish?
Model the risk matrix 8 min	Model likelihood x seriousness on a curated demo, placing a wrong-answer risk and a privacy risk.	Watch and annotate a copy of the matrix.	• How likely is this risk? • How serious would it be?
Final iteration 14 min	Support pairs to make their last planned improvement and re-test with known-answer cases.	Apply the final change and record before-and-after accuracy.	• Did the change help? • Did anything else break?
Risk review 12 min	Guide pairs to list risks, place them on the matrix, decide actions, and confirm no	Complete the risk matrix and write a safe-and-fair	• Which risk needs action now? • Is any private data present

	personal data is used.	statement.	anywhere?
Freeze and log 5 min	Have pairs freeze the version and complete the final iteration log for showcase.	Freeze the solution and finish the log.	• Is this the version we evaluated? • Is our evidence clear enough for others to trust?

Anticipated misconceptions

Misconception	How to address it
A rare risk is not worth checking.	A rare risk can still be very serious. Use the matrix: high seriousness needs a plan even at low likelihood.
If data has no names, it is automatically private-safe.	Other details can still identify someone. The safe rule is to keep all personal or identifying data out of prompts and training sets.
Once improved, we can keep changing it right up to showcase.	You freeze the evaluated version so the solution you present is the one you actually tested and judged safe.

Hands-on activity guide

Overview: Pairs complete a full final iteration on their signature solution, then run a structured risk review using a likelihood-by-seriousness matrix, confirm the solution is safe and fair, and freeze the evaluated version. This is the deeper, practical partner to the Core lesson.

Grouping / timing: Working pairs, with a cross-pair check for the risk matrix.

Tool: Printed risk matrix and final-iteration log; board demo; school-approved child-safe tool used by the teacher only for any live re-test. - Account-free and teacher-mediated. Students use paper matrices and curated outputs; no student logins and no personal data entered anywhere.

Facilitation steps:

1. Re-open the refinement log and set the Lab goal.
2. Make the final planned improvement to the solution.
3. Re-test with at least five known-answer cases and record before and after.
4. List possible risks (wrong answer, unfairness, privacy).
5. Place each risk on the likelihood-by-seriousness matrix.
6. Decide an action for any high risk and confirm no personal data is used.
7. Write a one-line safe-and-fair statement and freeze the version.
8. Swap matrices with another pair to sense-check placements.

Unplugged fallback: If no device or board is available, use simulated-AI cards and the paper matrix only: pairs judge curated outputs, plan the final change on paper, and complete the full risk review unplugged.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Self- and peer-evaluation against the tiered rubric (Emerging = expected)

• Check: Final iteration completed and a full risk review with a safe-and-fair statement

• Success criteria: Meets the Emerging criteria; completes a risk matrix, acts on one improvement, and confirms no personal data is used.

• Evidence: Final iteration log + completed risk matrix + safe-and-fair statement.

• Feedback / adjustment: Cross-pair sense-check plus teacher confirmation that risk judgements and privacy checks are sound.

Differentiation

Representation: Give the risk matrix as a labelled diagram, a spoken walkthrough and a worked example.

Action & expression: Learners may place risks with sticky notes, drawings or dictation.

Engagement: Use each pair's real user scenario so the risks feel authentic and worth managing.

People of Determination: Provide a pre-labelled matrix, chunked steps, sentence stems and extra time; allow a scribe.

EAL: Pre-teach likelihood, seriousness, fair, freeze with a bilingual glossary and icons; accept first-language notes.

Stretch / next tier: Ask pairs to propose a design change that would move a risk to a lower box and justify it.

Answer key / model responses

Task / item	Model response	Marking note
Place one risk on the matrix and explain.	A wrong answer is medium likelihood and medium seriousness, so I added a check rule to reduce it.	Look for both likelihood and seriousness reasoning.
State your final improvement with evidence.	I balanced my example data; accuracy rose from 4/6 to 6/6 with no new errors.	Requires before/after numbers.
Write a safe-and-fair statement.	My solution uses no personal data and treats all users the same, so it is safe and fair to use.	Must cover privacy and fairness.
Explain what freezing means.	I stopped changing it so the version I show is the one I tested and judged safe.	Accept any answer linking freeze to the evaluated version.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Completes a structured risk review	Places risks by likelihood and seriousness and decides actions.	Lists risks but placement or action is unclear.	No structured risk review.
Final iteration with evidence	Makes the final change and shows before-and-after test results.	Makes a change but evidence is thin.	No final iteration evidence.
Privacy and fairness confirmed	Confirms no personal data used and solution is fair.	Confirms one of privacy or fairness.	Neither confirmed.

Resources & links

- ADEK K-12 AI Fluency Framework - The four-strand outcomes this unit is built on. (https://www.adek.gov.ae)
- Risk matrix template - Printed likelihood-by-seriousness grid for the risk review. (school-shared)

Week 10 - Core | Showcase, Assessment & Reflection

Strand: Integration (Showcase, Assessment & Reflection) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Core)

Objectives & success criteria

- Present the finished solution and explain at least two design/evaluation choices.
- Demonstrate the term's key outcomes in a short assessment task.
- State one thing learned and set one goal for next term.

Student-facing success criteria:

- I can present my finished solution and explain at least two design or evaluation choices.
- I can show the term's key outcomes in a short assessment task.
- I can state one thing I learned and set one goal for next term.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: Across Term 1 you learned the difference between rule-based and learning systems, designed a signature solution, built it, evaluated it and, in Week 9, finished it with a risk review. You now have a frozen, safe-and-fair solution and a refinement log full of evidence.

Activate: Show the four-strand summary poster. Ask each learner to point to the strand they feel strongest in and say why.

Teach: Model a 60-second solution pitch that names the user, the problem, and two justified design/evaluation choices.

Apply: Learners draft two design/evaluation choices they will explain, with evidence.

Check: Cold-call three learners to state one of their two choices and its evidence.

Critical & analytical thinking focus

Evaluate & communicate - justify and reflect (Evaluate | DOK 3 | SOLO: Extended abstract (stretch))

The technical concept (teacher background)

Communicating a solution well is part of building it responsibly. A clear pitch names the real user, the problem, how the solution works, and the evidence that it is accurate, useful and safe. Explaining a design or evaluation choice means giving the reason behind a decision, not just describing what you did; for example, choosing a rule-based check because the task had a fixed correct answer, or balancing example data because one group was under-represented. A short performance assessment lets you show each of the four strands at the Emerging level: understanding concepts, designing and building, evaluating and using responsibly, and thinking about the wider effects. Reflection turns experience into learning by naming what worked, what was hard, and what you would do next. Setting a goal for next term, aimed at the next tier of Proficient, keeps your growth deliberate. Responsible communicators are honest about limits and never claim their solution is perfect.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Connect 6 min	Share the four-strand summary and the goal: present, be assessed, reflect. Remind students to be honest about limits.	Locate their two design/evaluation choices to explain.	• What are the four strands? • Which two choices will you justify?
Model the pitch 8 min	Model a 60-second pitch and a self-reflection sentence, thinking aloud about honesty and evidence.	Annotate the pitch structure and refine their own two choices.	• What makes a choice justified? • Where is my evidence?
Assessment task 15 min	Administer the short Term-1 performance task that samples all four strands; keep it individual and quiet.	Complete the assessment task, showing each strand at Emerging.	• Which strand does this question check? • Have I shown evidence?
Present and justify	Run brief pitches in small groups; ensure each learner explains two choices.	Present the finished solution and justify two	• Why did you make that choice? • How do you know it worked?

10 min		design/evaluation choices.	
Reflect and set a goal 6 min	Lead structured self-reflection and record each learner's next-tier goal for Term 2.	Write one thing learned and one goal for next term.	• What is your proudest outcome? • What is your goal toward Proficient?

Anticipated misconceptions

Misconception	How to address it
A good pitch means saying the solution is perfect.	Responsible presenting is honest about limits. Naming a limit and how you managed it shows stronger understanding.
Explaining a choice just means describing what I did.	Justifying a choice gives the reason behind it and the evidence, for example why a rule-based check fit the task.
The assessment is separate from my project.	The task samples the same four strands your project used, so your project evidence supports your answers.

Hands-on activity guide

Overview: Learners complete a short Term-1 performance task sampling all four strands at Emerging, then present their finished solution and justify two design/evaluation choices, and finish with a structured reflection and a Term-2 goal.

Grouping / timing: Individual for the assessment task; small groups of three or four for the pitches.

Tool: Printed assessment task and pitch card; four-strand summary poster; board demo only where needed, teacher-mediated. - Account-free. No student logins. The assessment and reflection are on paper; no personal data is entered into any tool.

Facilitation steps:

1. Read the four-strand summary and set out your evidence.
2. Complete the short Term-1 assessment task individually.
3. Prepare a 60-second pitch naming user, problem and two justified choices.
4. Present to your small group.
5. Give one peer a specific 'why' question about their choices.
6. Write one thing learned and one Term-2 goal toward Proficient.

Unplugged fallback: If groups cannot present live, learners record their pitch on the pitch card and swap cards for silent written peer feedback.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Summative Term-1 task samples all four strands at Emerging; reflection informs Term 2

• Check: Short Term-1 performance task marked with the analytic rubric + structured self-reflection

• Success criteria: Demonstrates each strand at Emerging; articulates two justified design/evaluation choices.

• Evidence: Assessed task + reflection; results reported per strand.

• Feedback / adjustment: Feedback flags each student's next tier target (to Proficient) for Term 2.

Differentiation

Representation: Offer the assessment in written and read-aloud forms and the pitch as a structured card.

Action & expression: Learners may present by speaking, using cue cards, or a recorded clip.

Engagement: Let learners choose which two choices to spotlight so the pitch feels personal.

People of Determination: Provide sentence stems, a pitch scaffold, extra time and a low-pressure presenting option.

EAL: Pre-teach present, justify, choice, goal with bilingual glossary; allow first-language rehearsal before the pitch.

Stretch / next tier: Ask learners to explain a trade-off between two choices, reaching Extended abstract thinking.

Answer key / model responses

Task / item	Model response	Marking note
Name your user and problem in one sentence.	My solution helps Omar in the canteen sort recycling faster and more accurately.	Must name a real user and a clear problem.
Justify one design choice.	I used a rule-based check because recycling categories have fixed correct answers.	Look for a reason, not just a description.
Justify one evaluation choice.	I tested with six known-answer cases because one lucky example is not enough evidence.	Should link to accuracy or fairness.
State one thing learned and one goal.	I learned to re-test after changes; my goal is to build a solution with balanced data next term.	Goal should aim toward Proficient.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Presents and justifies choices	Explains two design/evaluation choices with reasons and evidence.	Explains one choice, or reasons are partly clear.	Describes work without justifying choices.
Demonstrates four strands (assessment)	Shows each strand at Emerging in the task.	Shows most strands at Emerging.	Several strands not yet shown.
Reflection and goal	States one learning and one clear next-term goal.	States a learning or a goal, not both clearly.	No clear reflection or goal.

Resources & links

- ADEK K-12 AI Fluency Framework - The four-strand outcomes and tier descriptors used for reporting. (https://www.adek.gov.ae)
- Four-strand summary poster - Emerging-level summary of the term across all four strands. (school-shared)

Week 10 - Lab | Showcase Carousel and Reflection (Lab)

Strand: Integration (Showcase, Assessment & Reflection) | Framework ref(s): AIF·CU·G6·E AIF·SD·G6·E AIF·CE·G6·E AIF·GE·G6·E | 1 x 45-minute period (Lab)

Objectives & success criteria

- Present the solution in a showcase carousel and give structured peer feedback.
- Consolidate reflections into a Term-2 goal across the four strands.
- Confirm the responsible-use message for every solution on display.

Student-facing success criteria:

- I can present my solution in a showcase carousel and give structured peer feedback.
- I can consolidate my reflections into a Term-2 goal across the four strands.
- I can confirm my solution is responsible to use and honest about its limits.

Prior knowledge & Prerequisite Primer (10 min)

Assumes: In the Week 10 Core lesson you presented your solution, completed the Term-1 assessment, and drafted a reflection and goal. This Lab makes it hands-on: a showcase carousel where you rotate, present, and give structured peer feedback, then consolidate everything into one Term-2 goal across all four strands.

Activate: Learners set up their frozen solution and evidence at a station and read the peer-feedback stems.

Teach: Model 'two stars and a wish' feedback tied to the four strands on a curated demo station.

Apply: Learners practise one star and one wish on a partner's station before the carousel starts.

Check: Two learners share the feedback they drafted; teacher checks it is specific and kind.

Critical & analytical thinking focus

Evaluate & communicate - justify and reflect (Evaluate | DOK 3 | SOLO: Extended abstract (stretch))

The technical concept (teacher background)

A showcase carousel is a structured way for a whole class to present and give feedback in a short time: solutions stay at stations while learners rotate, present briefly, and leave written feedback. Good peer feedback is specific, kind and tied to the criteria, so 'two stars and a wish' works best when each comment names a strand and points to evidence. Consolidating reflections means gathering the small notes you made across the term into one clear picture of your growth across the four strands, then choosing a single goal for next term aimed at the Proficient tier. Every solution on display should carry a responsible-use message: it uses no personal data, it is honest about its limits, and it treats users fairly. A four-strand Emerging summary captures what a Grade 6 learner can now do across understanding concepts, designing and building, evaluating and using responsibly, and thinking about wider effects. Presenting and receiving feedback well is itself a fluency skill, because real builders improve through honest, respectful review.

Lesson plan

Phase | time	Teacher (I do / we do)	Students	Key questions (targeted thinking)
Connect 5 min	Explain the carousel rules and the feedback stems; reinforce the responsible-use message.	Set up their station with the frozen solution and evidence.	• What makes feedback specific and kind? • Is my responsible-use message on display?
Carousel round 1 12 min	Manage rotations; ensure presenters name two justified choices and visitors leave strand-tagged feedback.	Present at their station and give feedback at two others.	• Which strand does this comment address? • Where is the evidence?
Carousel round 2 10 min	Swap roles so all learners both present and review; prompt 'why' questions.	Give a second round of structured feedback and collect their own.	• Why did they make that choice? • What is one honest limit here?
Consolidate	Guide learners to gather feedback and	Complete the four-strand	• What can I now do in each

reflections 12 min	term notes into the four-strand summary and one Term-2 goal.	Emerging summary and set one goal.	strand? • What single goal moves me toward Proficient?
Close the term 6 min	Celebrate growth, confirm each responsible-use message, and collect the summaries for Term-2 planning.	Share one goal aloud and hand in the summary.	• What are you most proud of? • How will you keep your solutions responsible?

Anticipated misconceptions

Misconception	How to address it
Feedback should only say nice things.	Kind feedback still includes a wish. A specific, respectful wish tied to a strand helps a peer grow.
A goal can be vague, like 'do better'.	A strong goal is specific and tied to a strand, for example 'use balanced data to improve fairness'.
The responsible-use message is only for my project, not the showcase.	Every solution on display carries the message: no personal data, honest about limits, fair to users.

Hands-on activity guide

Overview: The class runs a showcase carousel: solutions stay at stations while learners rotate to present and leave structured, strand-tagged peer feedback. Learners then consolidate all feedback and term notes into a four-strand Emerging summary and one Term-2 goal, confirming each solution's responsible-use message.

Grouping / timing: Individual stations; whole-class rotation in two rounds; pairs support each other for consolidation.

Tool: Station cards, feedback slips, four-strand summary sheet; board demo only for the modelling step, teacher-mediated. - Account-free and paper-based. No student logins; no personal data on any station or in any tool.

Facilitation steps:

1. Set up your station with the frozen solution, evidence and responsible-use message.
2. Present your solution in 60 seconds, naming two justified choices.
3. Rotate to at least two other stations and leave 'two stars and a wish' feedback tagged to a strand.
4. Collect the feedback slips left at your own station.
5. Gather your term notes and feedback onto the four-strand summary.
6. Set one specific Term-2 goal aimed at Proficient.
7. Confirm your responsible-use message is complete and honest.

Unplugged fallback: If a live carousel is not possible, run a paper gallery: station cards and evidence are laid out, and learners circulate silently leaving written feedback slips, then consolidate as above.

Formative assessment (aligned to the objectives)

Formative check. • Targets: Summative Term-1 task samples all four strands at Emerging; reflection informs Term 2

• Check: Structured peer feedback across the carousel plus a completed four-strand summary and Term-2 goal

• Success criteria: Gives specific strand-tagged feedback, confirms responsible use, and sets one clear Term-2 goal.

• Evidence: Feedback slips + completed four-strand Emerging summary with goal.

• Feedback / adjustment: Peer feedback structured and strand-tagged; teacher flags each learner's next-tier target for Term 2.

Differentiation

Representation: Provide feedback stems, a modelled example and the summary as a labelled grid.

Action & expression: Learners may present live, with cue cards, or via a recorded clip at their station.

Engagement: Choice of which two choices to spotlight and a friendly rotation partner keep confidence high.

People of Determination: Offer a quieter rotation slot, pre-filled feedback stems, extra time, and a scribe for slips.

EAL: Pre-teach carousel, feedback, star, wish, consolidate with bilingual glossary; allow first-language rehearsal.

Stretch / next tier: Ask learners to synthesise feedback from several peers into one improved design idea for Term 2.

Answer key / model responses

Task / item	Model response	Marking note
Write one star tagged to a strand.	Star (Evaluate & use): you tested with six known-answer cases, which is strong evidence.	Must name a strand and evidence.
Write one wish tagged to a strand.	Wish (Wider effects): explain who might be treated unfairly and how you would prevent it.	Should be specific and kind.
State your four-strand summary in brief.	I can explain rule-based vs learning, build a solution, evaluate it for risk, and consider wider effects.	One clause per strand at Emerging.
Set one Term-2 goal.	Next term I will use balanced data so my solution is fairer to all users.	Specific and tied to a strand toward Proficient.

Assessment rubric (observable look-fors)

Criterion	Emerging (expected)	Approaching	Not yet
Structured peer feedback	Gives specific 'two stars and a wish' tagged to strands.	Gives feedback but it is general or untagged.	Little or unhelpful feedback.
Four-strand summary consolidated	Summarises all four strands at Emerging with evidence.	Summarises most strands.	Summary incomplete.
Term-2 goal and responsible use	Sets a specific goal and confirms responsible use.	Sets a goal or confirms responsible use, not both.	Neither is clear.

Resources & links

- ADEK K-12 AI Fluency Framework - The four-strand outcomes and tier descriptors for the summary. (https://www.adek.gov.ae)
- Carousel feedback slips and four-strand summary sheet - Templates for strand-tagged peer feedback and end-of-term consolidation. (school-shared)

Appendix A - Glossary of Key Terms

Teacher reference and student-facing definitions. Introduce each term where it first appears.

Data - information a system collects and uses to make a decision.

Label - a name or tag given to an example so a system can sort or learn it.

Rule - an instruction a system follows, often IF ... THEN ...

Outcome / output - the result a system produces after processing its input.

Fair / unfair - fair treats similar people or things equally; unfair wrongly favours or leaves out some.

Bias - a slant in a system's outputs, usually from unbalanced, skewed or low-quality data.

Personal information - data about a real person (name, photo, home, age) that must be kept private.

Rule-based system - a system that follows fixed IF-THEN rules a person wrote.

Learning system / model - a system that learns patterns from many examples (training data) instead of fixed rules.

Training data - the examples a learning system learns from.

Prompt - the words we give an AI system to tell it what to do.

Data quality / noisy data - how complete and correct data is; noisy data is messy or error-filled.

Trustworthy AI - an AI system whose outputs are reliable, fair and safe to use.

Appendix B - Understanding Bias: How Data Shapes Outputs

How learning systems use training data

At Grade 6 students meet learning systems (models). Unlike a rule-based system that follows fixed IF-THEN rules a person wrote, a learning system finds patterns in many examples called training data and uses those patterns to decide. So the training data shapes what the system does: if the examples are unbalanced, noisy or low quality, the model can be biased or wrong. This is why 'AI is just rules' is only half the story - modern AI also learns from data.

A concrete, age-appropriate example

The Animal-Sorter. We train a learning helper with many cat photos but very few dogs. Shown a new dog, it labels it 'cat'. Nobody wrote a wrong rule - the training data was unbalanced, so the model learned a biased pattern. Balanced, good-quality training data gives fairer, more accurate models.

Misconceptions to name and correct (say them aloud, then correct)

- 'AI is just fixed rules.' -> Rule-based systems follow rules; learning systems also learn patterns from training data. Grade 6 studies both.
- 'Data is neutral / always true.' -> Training data is chosen by people; unbalanced or noisy data biases the model.
- 'AI understands like a person.' -> A model matches learned patterns; it has no understanding.

Use as the core of the fairness/data-quality lesson and as a one-line check in every hands-on activity: 'What data did the system learn from? Who might be left out?'

Appendix C - Rule-Based vs Learning Systems (comparison)

	Rule-based system	**Learning system (model)**
How it decides	Follows fixed IF-THEN rules a person wrote.	Finds patterns in training data and uses them to decide.

Where it can go wrong	A wrong or unfair rule.	Unbalanced, noisy or low-quality training data.
Example	A sorting robot with IF-THEN cards.	An image labeller trained on many photos.
To improve it	Fix or rewrite the rule.	Improve the data (more balanced, cleaner).

Appendix D - Recommended Tools & Simulations + Unplugged Alternatives (UAE-accessible)

- Primary (teacher-mediated, account-free): Teachable Machine (teacher-run, no student logins) for a rule-vs-learning demo; Machine Learning for Kids (teacher-mediated); a prompt-based helper on the teacher screen.
- Fully unplugged core: printed clean-vs-noisy dataset cards, prompt-refinement logs and a rule-vs-learning comparison card - every lesson runs with no device.
- Alternatives for restricted environments: (1) paper-only clean-vs-noisy data sorts and prompt-refinement activities; (2) a single teacher-projected simulation or a pre-recorded demo played offline.
- Device requirements: one teacher device + projector is sufficient. Offline option: all demos can be pre-recorded; no student accounts or internet needed for the unplugged path. Confirm every tool against your school's approved-apps list (UAE).
- UDL notes travel with every lesson (multiple means of representation, action & expression, and engagement).

Appendix E - Criteria Checklist (accuracy, fairness, usefulness, reliability)

Use when judging any output. Print one prompt per card as 'criteria cards' for pairs.

- Accuracy: did the system give the right answer on the examples we tested?
- Coverage: did we test examples from every group?
- Fairness: does any group get the wrong output more often? Who might be left out?
- Usefulness: does the output actually help the task?
- Reliability: does it give consistent answers when we try again?
- Data check: was the data balanced, complete and clean? What was missing or noisy?
- Fix: what one change to the data, prompt or rule would improve it?

Appendix F - Privacy: Why We Protect Personal Information

Rule: we never put personal information (real names, photos, home address, age) into a system; we use synthetic (made-up) data only.

- Why - ethical: it is each person's right to keep their information private; sharing it can hurt or embarrass someone.
- Why - technical: once data enters a system it can be stored, copied and reused in ways we cannot take back, and small pieces of data can be combined to identify a person. Synthetic data removes this risk.

Appendix G - Printable Resources (templates)

Prompt frames

Sentence starters students use to instruct a helper: 'Please [task] using [these examples]. Show your steps. If unsure, say so.' Print as cards.

Clean vs noisy dataset cards

Example	Clean or noisy?	Problem to fix

Prompt-refinement log

Version	Prompt we tried	What the output missed	Improved prompt
1			
2			

Improvement / iteration log

Version	What we tested	What went wrong	Change we made	Better?
1				
2				

Appendix H - Real-World & Cross-Curricular Connections

- Real AI: data-and-model decisions power photo tagging, recommendation and image/voice recognition; helper prompting is how people use assistants and chat tools responsibly.
- UAE example: a school library or an app store that sorts and recommends by labels and past choices - a real 'uses data to decide' system students can see.
- Cross-curricular anchors: Maths (data, proportion, sets); Science (classification, evidence); Humanities / Moral education (fairness, privacy, responsible use); Literacy (clear prompts and success criteria).
- Deployment norms: real teams test a system for accuracy and fairness before people rely on it - exactly what the criteria checklist rehearses.

Appendix I - Looking Ahead

This unit prepares you for Grade 7 and beyond, where you build, test and evaluate learning systems more deeply.